

Soft Resilience in Hospitals Under Stress: Evidence from Healthcare-Associated Infections During COVID-19

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Abstract

This paper studies when communication-related soft resilience becomes more consequential for infection-control performance. I use pre-pandemic hospital communication quality as a proxy for this capacity. The analysis combines HCAHPS, CMS Hospital Compare, and Hospital Cost Report data for 3,366 U.S. hospitals from 2018 to 2021. I use the onset of COVID-19 as a common external stress shock. The empirical design examines whether infection outcomes changed differently after COVID-19 across hospitals with different pre-pandemic communication quality. The main relationship appears in BSI outcomes with higher COVID-induced diagnostic noise. A one-point increase in pre-pandemic communication quality is associated with about a 10 percent smaller post-COVID change in these infection rates relative to their sample means. No comparable pattern appears for non-BSI outcomes with lower COVID-induced diagnostic noise. I then examine two additional margins of effective task-related noise. The relationship is stronger among hospitals with less prior BSI-related experience. It also becomes stronger as local COVID-19 intensity rises for BSI outcomes with higher COVID-induced diagnostic noise. The results are robust to hospital fixed effects, pre-pandemic medical delivery quality, staffing, alternative communication measures, and sample restrictions. Together, the findings show that communication-related soft resilience is more strongly associated with infection-control performance when effective task-related noise is greater.

JEL codes: D23, I11, I18

Keywords: Communication-Related Soft Resilience, Hospital Resilience, Healthcare-Associated Infections, Diagnostic Noise, COVID-19

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1 Introduction

Crises expose how differently organizations can perform with similar resources. What may separate them is not only the resources they have, but also how effectively they use those resources when exceptional problems arise. Common problems can often be solved using existing knowledge. Exceptional problems, by contrast, require access to specialized knowledge and greater communication. Knowledge-based theories of organization formalize this distinction: common problems are handled where the relevant knowledge already resides, whereas more exceptional problems are referred to specialized problem solvers (Garicano, 2000). What remains unclear empirically is when the organizational capacity to meet these greater communication and problem-solving demands becomes consequential.

Hospitals during COVID-19 provide a useful setting to study this question because the pandemic increased diagnostic noise unevenly across infection-control tasks. Bloodstream infection (BSI) outcomes such as CLABSI and MRSA were particularly exposed: COVID-19 and BSI share systemic clinical signs, and the pandemic also complicated the interpretation of blood-culture results (Sacchetti et al., 2022; Cortes et al., 2023). In contrast, non-bloodstream infection (non-BSI) outcomes such as CAUTI and SSI are tied to more source-specific clinical contexts, including urinary catheterization and recent surgery. I therefore classify CLABSI and MRSA as BSI outcomes with higher COVID-induced diagnostic noise and CAUTI and SSI as non-BSI outcomes with lower COVID-induced diagnostic noise. This cross-task variation allows me to test whether the organizational capacity to meet greater communication and problem-solving demands becomes more consequential as diagnostic noise increases.

I use the term communication-related soft resilience to refer to hospitals’ organizational capacity for communication, responsiveness, and coordination. I proxy for communication-related soft resilience using pre-pandemic hospital communication quality from the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) survey. The measure combines ratings of nurse communication, doctor communication, staff responsiveness, and communication about medicines. For each hospital, I use the latest available pre-pandemic HCAHPS observation from 2018–2019. These measures directly capture patient-facing information exchange and staff responsiveness, while prior evidence suggests that they may also contain information about underlying coordination-related processes (von Knorring et al., 2020; Dobrzykowski and Tarafdar, 2015). I therefore interpret pre-pandemic hospital communication quality as an imperfect proxy for communication-related soft resilience, reflecting hospitals’ capacity for communication, responsiveness, and coordination.

The empirical analysis combines three data sources. HCAHPS provides hospital communication and patient-experience measures. CMS Hospital Compare provides quarterly infection outcomes. The Hospital Cost Report Information System provides hospital characteristics. The final sample contains 3,366 U.S. hospitals

observed from 2018 through 2021. I use the onset of COVID-19 as a common external stress shock. Pre-pandemic hospital communication quality serves as a predetermined proxy for communication-related soft resilience. The baseline difference-in-differences specification estimates whether infection outcomes changed differentially after COVID-19 across hospitals with different levels of pre-pandemic hospital communication quality. The key test compares BSI outcomes (higher COVID-induced diagnostic noise) and non-BSI outcomes (lower COVID-induced diagnostic noise). If communication-related soft resilience becomes more consequential as diagnostic noise increases, the differential post-COVID relationship should be stronger for BSI outcomes (higher COVID-induced diagnostic noise). The design therefore tests whether communication-related soft resilience becomes more consequential as diagnostic noise increases.

The main result is concentrated in BSI outcomes (higher COVID-induced diagnostic noise), including CLABSI and MRSA. A one-point increase in pre-pandemic hospital communication quality is associated with about a 10 percent smaller post-COVID change in CLABSI and MRSA rates relative to their sample means. No comparable negative relationship appears for the non-BSI outcomes (lower COVID-induced diagnostic noise), including CAUTI and surgical site infections. The same pattern for BSI outcomes (higher COVID-induced diagnostic noise) appears when pre-pandemic hospital communication quality is measured using a median split. Event-study estimates show no systematic differential pre-trends for either BSI outcomes (higher COVID-induced diagnostic noise) or non-BSI outcomes (lower COVID-induced diagnostic noise) before COVID-19. After the pandemic begins, the estimates become increasingly negative for the BSI outcomes (higher COVID-induced diagnostic noise), while no similar pattern emerges for the non-BSI outcomes (lower COVID-induced diagnostic noise).

I then test the same prediction along two additional margins of effective task-related noise. The first is prior BSI-related experience, proxied by pre-pandemic bacterial sepsis volume using SEP-1 reporting status. Hospitals are classified as Low_Volume if they have at least one missing SEP-1 report in the 2018–2019 pre-pandemic baseline sample. Pre-pandemic hospital communication quality is more strongly associated with smaller post-COVID changes in MRSA rates among hospitals classified as Low_Volume. The CLABSI estimate points in the same direction but is less precise. The second margin is local COVID-19 intensity. As local COVID-19 intensity rises, the negative relationship between pre-pandemic hospital communication quality and both BSI outcomes (higher COVID-induced diagnostic noise) becomes stronger. Neither pattern appears systematically among the non-BSI outcomes (lower COVID-induced diagnostic noise). Together, these results show a common pattern. Communication-related soft resilience becomes more consequential as effective task-related noise increases: across infection-control tasks with different levels of COVID-induced diagnostic noise, across hospitals with different levels of prior BSI-related experience, and across different levels of local COVID-19 intensity.

Several additional tests address alternative explanations. Hospital fixed effects control for time-invariant

hospital heterogeneity, and the estimates for BSI outcomes (higher COVID-induced diagnostic noise) remain largely unchanged. Controlling for pre-pandemic readmission performance addresses the possibility that pre-pandemic hospital communication quality reflects general medical delivery quality. Allowing post-pandemic outcomes to vary with staffing tests whether the results are driven by differences in staffing rather than communication-related soft resilience. I also compare communication with hospital quietness, another HCAHPS measure from the same survey system and the same pre-pandemic baseline observation. Quietness does not reproduce the negative pattern for BSI outcomes (higher COVID-induced diagnostic noise), reducing the concern that the results reflect general patient experience rather than communication-related soft resilience. Replacing the composite measure with nurse communication, doctor communication, staff responsiveness, and communication about medicines shows that the pattern for BSI outcomes (higher COVID-induced diagnostic noise) is not driven by a single communication component. I further exclude hospitals in the middle of the communication distribution and retain only hospitals with relatively high or low communication scores. The results remain similar, showing that the baseline pattern is not driven by small differences around the center of the communication distribution. Together, these tests make explanations based on time-invariant hospital heterogeneity, general medical delivery quality, staffing, general patient experience, or a single communication component less consistent with the evidence.

The paper first contributes to organizational economics on knowledge transmission and communication inside organizations. [Garicano \(2000\)](#) develops a theory in which production workers acquire knowledge about common or easy problems, while more exceptional or harder problems are handled by specialized problem solvers. Communication allows for the specialized acquisition of knowledge, while a knowledge-based hierarchy organizes the matching of problems with those who know how to solve them. A growing empirical literature examines how communication and knowledge flows affect worker and team performance. In a field experiment, [Sandvik et al. \(2020\)](#) show that reducing frictions to knowledge seeking increases worker productivity. [Battiston et al. \(2021\)](#) show that the second worker in a sequential production process works faster when face-to-face communication with the first worker is possible, with a larger effect when the second worker is busier. [Emanuel et al. \(2026\)](#) find that sitting near teammates increases coding feedback and improves code quality, with the gains concentrated among younger and less-tenured employees. [Lyons \(2017\)](#) show that team organization improves outcomes when workers are from the same country but worsens outcomes when workers are nationally diverse; these differences are more pronounced among teams with specialized skills, and further evidence suggests that nationally diverse teams face communication difficulties. I complement this literature by studying when communication-related soft resilience matters more for infection-control performance. I use pre-pandemic hospital communication quality as its empirical proxy. Its relationship with infection-control performance is stronger when effective task-related noise is higher. I find this pattern when COVID-induced diagnostic noise is higher, prior BSI-related experience is lower, and local COVID-19 intensity is higher.

Second, the paper contributes to the economics literature on healthcare production and provider performance under stress. [Bloom et al. \(2015\)](#) show that greater competition among public hospitals improves management quality and hospital performance, including survival following emergency heart-attack admissions. [Chandra et al. \(2016\)](#) show that higher-quality hospitals have larger market shares and are more likely to expand over time. A related health-economics literature studies provider performance during the COVID-19 shock. [Cronin and Evans \(2022\)](#) find that higher-quality nursing homes had lower COVID-19 mortality and less resident spread conditional on the introduction of COVID-19 into the facility. In the hospital setting, [Fetzer et al. \(2024\)](#) identify COVID-19 pressures as a key driver of excess mortality among non-COVID-19 patients in the English NHS. More broadly, quantitative measures of hospital resilience have focused heavily on routine clinical activity and input capacity, particularly beds, staff, and equipment, while organizational and intangible factors remain difficult to translate into objective quantitative measures ([Cavaliere et al., 2025](#)). I add to this literature by studying communication-related soft resilience in hospitals. I use pre-pandemic hospital communication quality as its empirical proxy. This measure complements existing hospital-resilience measures that focus primarily on clinical activity and physical resources.

The remainder of the paper is organized as follows. Section 2 develops the conceptual framework and the main prediction linking communication-related soft resilience to diagnostic noise. Section 3 describes the data, measures, and empirical strategy. Section 4 presents the baseline evidence, examines alternative explanations and heterogeneity, and reports robustness tests. Section 5 interprets the findings and discusses the main limitations. Section 6 concludes.

2 Conceptual Framework

This section develops a framework of hospital resilience during public health crises. I distinguish between hard resilience, which reflects tangible hospital capacity, and soft resilience, which reflects organizational capacity to process information, coordinate actions, and adapt when routine conditions are disrupted. I then explain why pre-pandemic hospital communication quality can serve as an empirical proxy for communication-related soft resilience. Finally, I develop the prediction that communication-related soft resilience should become more consequential for infection-control tasks exposed to greater COVID-induced diagnostic noise.

2.1 Hospital Resilience and Infection Control

Hospital resilience has two complementary dimensions: hard resilience and soft resilience. Hard resilience reflects the structural and material capacity that allows hospitals to maintain operations under stress. Soft resilience reflects the organizational capacity to process information, coordinate actions, and adapt when routine conditions are disrupted.

Hard resilience includes hospital infrastructure, medical equipment, and other physical capacities needed

to sustain healthcare delivery during a crisis (Khalil et al., 2022; Cavalieri et al., 2025). These resources provide the material foundation for infection-control activities. Physical capacity affects hospitals’ ability to isolate patients, expand treatment capacity, and maintain essential clinical services when demand rises. Resource constraints such as inadequate infrastructure, overcrowding, and shortages of critical equipment can directly impede infection prevention (Alp et al., 2011). Hard resilience therefore determines whether hospitals possess the physical capacity needed to respond to a health shock.

Soft resilience captures a different dimension of hospital capacity. The hospital-resilience literature distinguishes these organizational capacities from structural and material resilience, emphasizing leadership and coordination, communication and information systems, and adaptive organizational processes as important components of soft resilience (Khalil et al., 2022).

These organizational capacities are directly relevant to infection control. Effective infection prevention depends not only on physical resources and formal protocols, but also on the organizational processes through which prevention practices are implemented across clinical staff. Systematic reviews identify hospital-level organization, multidisciplinary prevention programs, organizational culture, communication, and collaboration as important correlates of stronger infection-prevention practices and lower healthcare-associated infection rates (Zingg et al., 2015; De Bono et al., 2014; van Buijtene and Foster, 2019). Communication itself can also affect implementation: interventions that improved the transmission of infection-control information during patient transfers substantially increased adherence to infection-control precautions (Ong et al., 2013). Although this evidence is not uniformly causal, it indicates that hospitals differ in their ability to translate available resources and formal infection-control practices into effective implementation. Soft resilience therefore captures the organizational capacity through which hospitals process information, coordinate the use of available resources, and sustain effective clinical responses under stress.

Soft resilience is multidimensional and cannot be observed directly in the available data. I therefore use pre-pandemic hospital communication quality as a proxy for communication-related soft resilience. The empirical measure combines patients’ assessments of nurse communication, doctor communication, staff responsiveness, and communication about medicines. These measures directly capture patient-facing information exchange and staff responsiveness. They do not directly measure soft resilience as a whole or other dimensions such as leadership and organizational adaptation.

Patient-facing communication may nevertheless contain information about underlying coordination-related processes. von Knorring et al. (2020) show that stronger nurse–physician teamwork is associated with greater consistency in the communication patients receive from hospital staff. Using hospital-level organizational data matched to CMS HCAHPS measures, Dobrzykowski and Tarafdar (2015) similarly show that stronger information exchange among physicians and hospital staff is associated with better provider–patient communication. Together, these findings suggest that patient-facing communication directly captures infor-

mation exchange and responsiveness while also providing an imperfect signal of the coordination processes underlying care delivery. I therefore interpret pre-pandemic hospital communication quality as a proxy for communication-related soft resilience, capturing hospitals' capacity for communication, responsiveness, and coordination rather than soft resilience in its entirety. Communication is not treated as the sole organizational mechanism affecting infection outcomes. Instead, pre-pandemic hospital communication quality provides an observable proxy for communication-related soft resilience. The next section develops the prediction that communication-related soft resilience should become more consequential as diagnostic noise increases.

2.2 Hypothesis Development

Healthcare-associated infections (HAIs) differ in their clinical presentation, diagnostic pathways, and management (Monegro et al., 2017). I use these differences to characterize the diagnostic noise hospitals face when identifying an infection and its likely source. I focus on five HAI outcomes: central line-associated bloodstream infection (CLABSI; BSI), Methicillin-resistant *Staphylococcus aureus* bloodstream infection (MRSA; BSI), catheter-associated urinary tract infection (CAUTI; non-BSI), surgical site infection following colon surgery (SSI-CS; non-BSI), and surgical site infection following abdominal hysterectomy (SSI-AH; non-BSI).

COVID-19 increased diagnostic noise particularly for BSI outcomes. COVID-19 and BSI can share systemic clinical signs, making the presence and source of infection more difficult to distinguish. Blood-culture contamination also increased during the pandemic, complicating the interpretation of blood-culture results used to identify BSI (Sacchetti et al., 2022). This problem is directly relevant to CLABSI (BSI). Fakhil et al. (2022) note that CLABSI classification can be complicated in severely ill COVID-19 patients because surveillance criteria rely partly on systemic clinical signs and blood-culture results. The same concern applies to MRSA (BSI). MRSA (BSI) increased sharply among hospitalized patients with COVID-19, and evidence from this setting suggests that central-line-associated BSI may have been underestimated (Aslam et al., 2023). More broadly, clinical studies during the pandemic document difficulty distinguishing BSI from COVID-19 and delays in BSI treatment when clinical attention was directed toward excluding COVID-19 (Cortes et al., 2023; Miyagami et al., 2021). Together, this evidence indicates that COVID-19 increased diagnostic noise for BSI outcomes.

CAUTI (non-BSI) and SSI (non-BSI) provide useful comparison outcomes because their identification is tied to more source-specific clinical contexts. For CAUTI (non-BSI)¹, urinary catheterization and urinary-tract evidence provide information about the likely infection source. Fakhil et al. (2022) document a sharp increase in CLABSI (BSI) during COVID-19 but no comparable increase in CAUTI (non-BSI), despite increased use of both central lines and urinary catheters. They also note that CLABSI (BSI) and CAUTI (non-BSI) differ in their surveillance and identification pathways.

¹<https://www.cdc.gov/nhsn/psc/uti/index.html>

SSI (non-BSI)² is similarly tied to a specific clinical context: a recent surgical procedure and the corresponding operative site. National NHSN data show no significant increase in SSI-CS (non-BSI) or SSI-AH (non-BSI) during 2020, while CLABSI (BSI) and MRSA (BSI) increased substantially (Weiner-Lastinger et al., 2022). One possible explanation is that colon surgery and abdominal hysterectomy were not typically part of COVID-19 care and that operating-room process flows remained relatively stable during the pandemic (Lastinger et al., 2023).

These source-specific contexts do not eliminate diagnostic noise, but they reduce the additional diagnostic noise introduced by COVID-19. I therefore classify CLABSI (BSI) and MRSA (BSI) as outcomes with higher COVID-induced diagnostic noise and CAUTI (non-BSI), SSI-CS (non-BSI), and SSI-AH (non-BSI) as outcomes with lower COVID-induced diagnostic noise. This distinction is relative: lower COVID-induced diagnostic noise means that COVID-19 added less uncertainty about the infection source, not that CAUTI (non-BSI) or SSI (non-BSI) is easy to diagnose.

This difference in diagnostic noise generates the main empirical prediction. When diagnostic noise is lower, hospitals can rely more heavily on existing knowledge. Differences in communication-related soft resilience should therefore be less consequential. As diagnostic noise increases, hospitals face greater demands for interpreting information, communicating concerns, and coordinating further evaluation across clinical staff. Communication-related soft resilience should therefore become more consequential as diagnostic noise increases. This prediction follows the broader logic of organizational economics. Garicano (2000) distinguishes common problems that can be handled using existing knowledge from more exceptional problems that require specialized problem solving and greater communication. In the hospital setting, higher diagnostic noise makes infection-control problems more exceptional in this sense: existing information becomes less sufficient for identifying the infection source, increasing the need for problem solving and communication. Communication-related soft resilience should therefore become more consequential as diagnostic noise increases.

Because communication-related soft resilience is not directly observable, I use pre-pandemic hospital communication quality as its empirical proxy. As discussed above, the measure directly captures patient-facing information exchange and responsiveness and may also contain information about underlying coordination processes. The prediction is therefore not that pre-pandemic hospital communication quality should be uniformly associated with better infection outcomes. Rather, its relationship with infection outcomes should be stronger for outcomes exposed to greater COVID-induced diagnostic noise.

Hypothesis 1. *During the COVID-19 pandemic, hospitals with greater communication-related soft resilience, proxied by pre-pandemic hospital communication quality, should experience relatively better outcomes for infections exposed to greater COVID-induced diagnostic noise. This relationship should be stronger for*

²<https://www.cdc.gov/nhsn/psc/ssi/index.html>

BSI outcomes with higher COVID-induced diagnostic noise, including CLABSI and MRSA, than for non-BSI outcomes with lower COVID-induced diagnostic noise, including CAUTI and SSI.

3 Data and Empirical Strategy

3.1 Data Sources

I combine three datasets from the Centers for Medicare & Medicaid Services (CMS), covering 2018 through 2021³. CMS Hospital Compare provides hospital-level infection measures. The main outcomes are central line-associated bloodstream infection (CLABSI), catheter-associated urinary tract infection (CAUTI), methicillin-resistant *Staphylococcus aureus* (MRSA) bloodstream infection, and surgical site infections (SSI) following colon surgery and abdominal hysterectomy. The Hospital Cost Report Information System (HCRIS) provides annual hospital characteristics, including staffing, bed capacity, costs, and revenues. The Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) survey provides standardized hospital-level measures of patient experience.

The HCAHPS survey was developed by CMS and the Agency for Healthcare Research and Quality (AHRQ). It is administered year-round to a random sample of adult patients discharged from medical, surgical, and maternity service lines. The survey is not limited to Medicare beneficiaries. It contains 29 questions, including 19 substantive patient-experience items, 3 screening questions, and 7 demographic questions used for patient-mix adjustment. CMS reports the results in 6 composite topics, 2 individual topics, and 2 global topics. The composite topics include nurse communication, doctor communication, staff responsiveness, communication about medicines, discharge information, and care transition. The individual topics measure hospital cleanliness and quietness. The global topics measure overall hospital rating and willingness to recommend the hospital.

HCAHPS data are collected by organizations trained in standardized HCAHPS procedures. The survey covers more than 4,000 short-term, acute-care, non-specialty hospitals, including hospitals operated by the Veterans Health Administration and the Department of Defense. Hospitals aim to complete at least 300 surveys annually. CMS reports hospital results four times per year using data from the preceding rolling year. CMS cleans and adjusts the data before public release.

The initial Hospital Compare sample contains 4,904 hospitals. Requiring infection data both before and after the onset of COVID-19 reduces the sample to 4,721 hospitals. Requiring HCAHPS communication measures further reduces the sample to 3,366 hospitals. I assign each observation to the endpoint of its reporting window.

³<https://data.cms.gov>

The infection data are reported in overlapping measurement windows. The five pre-pandemic windows are 2018.01–2018.12, 2018.04–2019.03, 2018.07–2019.06, 2018.10–2019.09, and 2018.12–2019.12. The five windows ending after the onset of COVID-19 are 2019.04–2020.09, 2019.07–2020.12, 2019.10–2021.03, 2020.10–2021.09, and 2021.01–2021.12. The empirical analysis uses the endpoint of each measurement window to define time.

Table 3 in the appendix compares hospitals included in and excluded from the final sample. Excluded hospitals tend to be smaller and have lower staffing levels and fewer financial resources. Their reported infection outcomes are also more dispersed. The final estimation sample is therefore selected toward hospitals with more complete communication and infection reporting. This selection may limit the representativeness of the sample for the broader U.S. hospital population.

3.2 Measures and Summary Statistics

3.2.1 Healthcare-Associated Infection Rates

The main analysis uses five healthcare-associated infection (HAI) outcomes: CLABSI, MRSA bloodstream infection, CAUTI, SSI following colon surgery (SSI-CS), and SSI following abdominal hysterectomy (SSI-AH).⁴ To limit the influence of extreme observations, I winsorize CLABSI, CAUTI, SSI-CS, and SSI-AH at the 99th percentile. MRSA is winsorized at the 99.9th percentile.

The CLABSI rate is the number of CLABSI events divided by central-line device days and is expressed per 1,000 device days. Central-line device days measure the total number of days during which hospitalized patients had a central venous catheter in place.

$$CLABSI\ Rate = \frac{CLABSI\ Number}{Central\ Line\ Device\ Days} * 1000 \quad (1)$$

The CAUTI rate is the number of CAUTI events divided by urinary-catheter device days and is expressed per 1,000 device days. Urinary-catheter device days measure the total number of days during which hospitalized patients had an indwelling urinary catheter in place.

$$CAUTI\ Rate = \frac{CAUTI\ Number}{Urinary\ Catheter\ Device\ Days} * 1000 \quad (2)$$

The SSI-CS rate is the number of observed surgical site infections following colon surgery divided by the number of colon-surgery procedures. It is expressed per 100 procedures.

⁴The infection-rate measures follow the methodology described by the Infectious Disease Epidemiology Section of the Louisiana Department of Health. <https://ldh.la.gov/assets/oph/Center-PHCH/Center-CH/infectious-epi/HAI/HAIworkshop2017/handoutsD1/IC-StatsHAIa.pdf>

$$SSI_{CS} \text{ Rate} = \frac{SSI_{CS} \text{ Number}}{\text{Colon Surgery Procedures}} * 100 \quad (3)$$

The SSI-AH rate is the number of observed surgical site infections following abdominal hysterectomy divided by the number of abdominal-hysterectomy procedures. It is expressed per 100 procedures.

$$SSI_{AH} \text{ Rate} = \frac{SSI_{AH} \text{ Number}}{\text{Abdominal Hysterectomy Surgery Procedures}} * 100 \quad (4)$$

The MRSA bloodstream infection rate is the number of MRSA bloodstream infection cases divided by patient days and is expressed per 1,000 patient days. Patient days measure the total number of inpatient bed-days accumulated across hospitalized patients during the reporting period.

$$MRSA \text{ Rate} = \frac{MRSA \text{ Number}}{\text{Number of Patient Days}} * 1000 \quad (5)$$

3.2.2 Hospital Communication Measure

I use HCAHPS communication ratings to construct pre-pandemic hospital communication quality. HCAHPS reports four relevant hospital-level measures on a 1–5 star scale: Nurse Communication, Doctor Communication, Staff Responsiveness, and Communication about Medicines.

These measures capture related but distinct aspects of patient-facing information exchange and responsiveness. They are standardized across hospitals and allow consistent cross-hospital comparisons. At the same time, they are patient-reported measures and may reflect differences in individual perceptions or recall. They also capture only selected dimensions of hospital communication. They therefore do not directly measure communication-related soft resilience.

The individual HCAHPS communication measures also differ in their relationship with broader hospital quality. For example, nurse communication is more strongly correlated with overall hospital quality than doctor communication (Belasen et al., 2021). Relying on a single component may therefore place disproportionate weight on one dimension of patient experience. I address this concern by combining the four measures into a composite measure.

Pre-pandemic hospital communication quality is defined as the average of Nurse Communication, Doctor Communication, Staff Responsiveness, and Communication about Medicines:

$$\text{Communication} = \frac{\text{Nurse} + \text{Doctor} + \text{Staff} + \text{Medicines}}{4} \quad (6)$$

where Nurse denotes the Nurse Communication star rating, Doctor denotes the Doctor Communication star rating, Staff denotes the Staff Responsiveness star rating, and Medicines denotes the Communication about Medicines star rating.

For each hospital, I define `Communication_2019` using its latest available pre-pandemic HCAHPS observation from 2018–2019. I use the latest available observation because communication-related soft resilience may change over time. An earlier measurement window may therefore be less representative of a hospital’s capacity immediately before COVID-19. This construction captures the most recent available measure of that capacity while keeping it predetermined with respect to the pandemic. `Communication_2019` is fixed throughout the analysis. The composite measure reduces reliance on any single communication component. I also examine each component separately in the robustness analysis. These tests assess whether the baseline results depend on the construction of the composite measure.

3.2.3 Additional Variables

The analysis includes additional measures of patient experience, hospital resources, staffing, and local COVID-19 conditions. The baseline patient-experience measures include cleanliness, quietness, willingness to recommend, High Rating, and Hospital Summary Rating. All are drawn from HCAHPS. For each hospital, I use the same latest available pre-pandemic observation from 2018–2019 that is used to construct pre-pandemic hospital communication quality. Cleanliness and quietness capture dimensions of the physical hospital environment. Willingness to recommend, High Rating, and Hospital Summary Rating capture broader assessments of hospital experience. Cleanliness is directly related to infection prevention ([Bogusz et al., 2013](#); [Ramphal et al., 2014](#)). Quietness is used later as a comparison measure. It is collected through the same HCAHPS framework as pre-pandemic hospital communication quality but captures a conceptually distinct dimension of patient experience. These controls help distinguish pre-pandemic hospital communication quality from more general differences in patient-reported hospital quality.

The baseline hospital resource and staffing measures are constructed from HCRIS. The resource measures include total cost, total operating expenses, inpatient hospital revenue, inpatient total patient revenue, bed capacity, inpatient bed days, and available bed days. HCRIS information is matched to hospitals by hospital and year. I then retain the values associated with the same baseline hospital observation used to construct pre-pandemic hospital communication quality. Staff Number is constructed in the same way. These variables capture differences in hospital scale, available resources, and staffing capacity. Resource constraints can affect hospitals’ ability to maintain infection-control practices ([Upadhyay and Smith, 2023](#)). Inadequate staffing has also been linked to higher HAI rates ([Stone et al., 2007](#)). I use time-varying staffing measures from the annual HCRIS data in the alternative-explanation analysis.

Finally, I use county-level COVID-19 cases and deaths to measure local pandemic severity. I first obtain

cumulative cases and deaths at the end of each county-quarter. I then compute quarterly changes in these values to obtain new cases and deaths. The county-level measures are matched to hospitals using county FIPS codes derived from hospital ZIP codes. COVID-19 cases are expressed per 100,000 people, and deaths are expressed per 1,000 people. These variables vary over time and capture differences in local COVID-19 exposure that may affect hospitals’ operating environments (Barowskin, 2023). A detailed definition of each variable is provided in Table 1.

3.2.4 Summary Statistics

Table 2 reports summary statistics for the sample used in the main empirical analysis. The final sample contains 3,366 hospitals observed from 2018 through 2021. Reporting availability differs across infection measures. The number of non-missing hospital-window observations therefore varies across outcomes. The table reports the number of observations, mean, standard deviation, and maximum for each variable.

[Insert Table 2]

The mean CLABSI rate is 0.60 per 1,000 central-line device days, with a standard deviation of 0.74. The mean CAUTI rate is 0.69 per 1,000 urinary-catheter device days, also with a standard deviation of 0.74. The mean SSI-CS rate is 2.05 per 100 colon-surgery procedures, with a standard deviation of 2.81. The mean SSI-AH rate is 0.70 per 100 abdominal-hysterectomy procedures, with a standard deviation of 1.48. The MRSA bloodstream infection rate averages 0.04 per 1,000 patient days, with a standard deviation of 0.06.

The annual statistics also show changes in infection outcomes over the sample period. The mean CLABSI rate increases from 0.49 in 2019 to 0.61 in 2020 and 0.76 in 2021. The mean SSI-CS rate declines modestly from 2.08 in 2019 to 1.99 in 2021. These are descriptive changes in infection outcomes. They do not identify the role of pre-pandemic hospital communication quality.

The HCAHPS measures are reported on a 1–5 scale. Pre-pandemic hospital communication quality has a sample mean of 3.23 and a standard deviation of 0.89. Cleanliness averages 2.89, quietness 3.20, willingness to recommend 3.18, High Rating 3.29, and Hospital Summary Rating 3.23. Among the four communication components, Nurse Communication has the highest mean at 3.50. Staff Responsiveness averages 3.28, Doctor Communication 3.09, and Communication about Medicines 3.07. All baseline HCAHPS measures use each hospital’s latest available pre-pandemic observation from 2018–2019. They therefore capture predetermined cross-hospital differences rather than contemporaneous responses to COVID-19.

Hospital financial and staffing characteristics also vary substantially across the sample. Total cost averages \$0.29 billion, with a standard deviation of \$0.45 billion and a maximum of \$6.16 billion. Total operating expenses also average approximately \$0.29 billion. Inpatient hospital revenue averages \$0.12 billion. The baseline staff measure averages 1.28 when expressed in thousands, with a standard deviation of 1.83. The

time-varying FTE staffing measure averages 0.13 when expressed in thousands, with a standard deviation of 0.19. These statistics show substantial variation in hospital scale and resources across the sample.

The table also reports county-level COVID-19 exposure. New COVID-19 cases average 0.12 per 100,000 people over the full sample, while new deaths average 0.15 per 1,000 people. Both variables equal zero in the pre-pandemic period and increase beginning in 2020. I later combine cases and deaths into a local COVID-19 intensity index using their normalized ranks. The index has a mean of 0.500 and a standard deviation of 0.267 in the full sample. Average values are higher in 2020 and 2021.

Finally, the SEP-1 measure has a mean of 0.17, and the hospital-wide 30-day readmission rate averages 0.15. SEP-1 information is used in the heterogeneity analysis to construct an imperfect proxy for pre-pandemic bacterial sepsis volume and prior BSI-related experience. The readmission measure is used to examine whether the baseline relationship can be explained by differences in general medical delivery quality.

[Insert Table 3]

Table 3 compares the 3,366 hospitals in the estimation sample with the 1,355 hospitals excluded because of missing communication data or incomplete infection reporting. Excluded hospitals have lower staffing levels. Their average staff measure is 0.29 when expressed in thousands. They also have smaller financial scale, with average total costs of approximately \$0.02 billion. Their reported infection outcomes are more dispersed. For example, the standard deviation of the CLABSI rate is 2.49 among excluded hospitals, compared with 0.74 in the estimation sample. CAUTI and SSI rates also show greater dispersion among excluded hospitals.

These differences show that excluded hospitals are smaller and less well resourced on average and have more variable reported infection outcomes. They do not establish why reporting is incomplete. They also do not establish that excluded hospitals provide lower-quality care. The comparison instead highlights a sample-selection limitation. The final sample is weighted toward hospitals with more complete HCAHPS and infection reporting and may not represent the full population of U.S. hospitals.

3.3 Identification Strategy

To test Hypothesis 1, I use the onset of COVID-19 to define the post period. I examine whether hospital infection outcomes changed differently with pre-pandemic hospital communication quality. The empirical design compares hospitals with different levels of pre-pandemic hospital communication quality and tracks their infection outcomes before and after the onset of COVID-19.

Pre-pandemic hospital communication quality is fixed before COVID-19. For each hospital, I use the latest available observation from the 2018–2019 pre-pandemic baseline sample. The same baseline observation is used to construct the other HCAHPS patient-experience controls and the corresponding HCRIS resource and

staffing measures. This approach uses hospital characteristics measured as close as possible to COVID-19 while keeping them predetermined.

I estimate two main specifications. The first uses pre-pandemic hospital communication quality continuously on its 1–5 scale. The second uses a median split. I define $H_j = 1$ for hospitals with pre-pandemic hospital communication quality above 3.25 and $H_j = 0$ otherwise. The baseline specification is

$$Y_{j,t} = \alpha + \beta_1 T_t \times H_j + \beta_2 H_j + \beta_3 T_t + X'_{j,t} \gamma + \lambda_t + \delta_s + \epsilon_{j,t}, \quad (7)$$

where j indexes hospitals and t indexes the endpoint of each rolling measurement window. The estimation sample runs from 2018.Q4 through 2021.Q4. $Y_{j,t}$ denotes the hospital infection outcome. The five outcomes are CLABSI, MRSA bloodstream infection, CAUTI, SSI-CS, and SSI-AH. As defined in Section 2, CLABSI and MRSA are BSI outcomes with higher COVID-induced diagnostic noise. CAUTI, SSI-CS, and SSI-AH are non-BSI outcomes with lower COVID-induced diagnostic noise.

T_t is an indicator equal to one for measurement windows ending after March 2020, corresponding to 2020.Q2 through 2021.Q4. It equals zero for measurement windows ending from 2018.Q4 through 2020.Q1. In the continuous specification, H_j denotes pre-pandemic hospital communication quality. In the median-split specification, H_j denotes the above-median indicator. Because year-quarter fixed effects are included, the main effect of T_t is absorbed and is not separately estimated.

The vector $X_{j,t}$ includes baseline patient-experience controls: Hospital Summary Rating, Clean, Quiet, Recommend, and High Rating. It also includes baseline HCRIS measures of Total Cost, Total Operating Expenses, Inpatient Hospital Revenue, and Staff Number. These baseline controls are fixed using the same pre-pandemic hospital observation described above. I further control for time-varying county-level COVID-19 cases and deaths to account for differences in local pandemic severity.

All specifications include year-quarter fixed effects, λ_t , and state fixed effects, δ_s . Year-quarter fixed effects absorb shocks common to hospitals in a given period. State fixed effects absorb persistent differences across states. Standard errors are clustered at the state level to account for within-state correlation and serial correlation from the repeated and overlapping measurement windows (Bertrand et al., 2004).

For the median-split specification, I also estimate an event-study DID using the above- and below-median pre-pandemic hospital communication quality groups. The omitted reference period is 2019.Q4. I use the pre-period coefficients to assess differential pre-trends. The remaining coefficients describe how the difference between the two groups evolves around the onset of COVID-19.

The coefficient of interest is β_1 . In the continuous specification, it measures how the post-COVID change

in infection rates varies with a one-point increase in pre-pandemic hospital communication quality. In the median-split specification, it measures the difference in the post-COVID change between hospitals above and at or below the median. Hypothesis 1 predicts $\beta_1 < 0$ for BSI outcomes with higher COVID-induced diagnostic noise. No clear and consistent directional pattern is predicted for non-BSI outcomes with lower COVID-induced diagnostic noise.

4 Empirical Evidence

4.1 Baseline Evidence on Communication-Related Soft Resilience

This section presents the baseline evidence on whether pre-pandemic hospital communication quality predicts differential changes in hospital infection outcomes during COVID-19. I first use the continuous measure of pre-pandemic hospital communication quality and then report a median-split specification that provides a more transparent comparison between hospitals with relatively high and low pre-pandemic hospital communication quality. Finally, I estimate an event-study version of the median-split DID specification to assess differential pre-trends and examine the dynamic evolution of the estimated relationships around the onset of COVID-19.

4.1.1 Continuous DID Estimates

Table 4 reports the baseline estimates using the continuous measure of pre-pandemic hospital communication quality. The coefficient on $\text{Post} \times \text{Communication}_{2019}$ is negative and statistically significant for both BSI outcomes (higher COVID-induced diagnostic noise). A one-point higher pre-pandemic hospital communication score is associated with a 0.063 smaller post-COVID change in the CLABSI rate and a 0.004 smaller post-COVID change in the MRSA rate. Relative to their respective sample means, these estimates correspond to approximately 10 percent for both outcomes.

[Insert Table 4]

In contrast, the interaction coefficients for CAUTI, SSI-CS, and SSI-AH are statistically indistinguishable from zero. The resulting pattern is therefore concentrated in BSI outcomes (higher COVID-induced diagnostic noise) rather than non-BSI outcomes (lower COVID-induced diagnostic noise). This distinction is consistent with Hypothesis 1: hospitals with higher pre-pandemic hospital communication quality experienced smaller post-COVID changes in infection rates primarily for outcomes involving greater COVID-induced diagnostic noise.

4.1.2 Median-Split DID Estimates

Table 5 presents the corresponding estimates after dividing hospitals into above-median and at-or-below-median groups based on pre-pandemic hospital communication quality. The results closely mirror those from the continuous specification. Hospitals with above-median pre-pandemic hospital communication quality experienced a 0.086 smaller post-COVID change in the CLABSI rate and a 0.004 smaller post-COVID change in the MRSA rate relative to hospitals at or below the median. These estimates correspond to approximately 14 percent and 10 percent of the respective sample means.

[Insert Table 5]

Again, there is no comparable differential change for CAUTI, SSI-CS, or SSI-AH. The median-split specification therefore produces the same cross-outcome pattern as the continuous specification: the association between pre-pandemic hospital communication quality and post-COVID infection outcomes is concentrated in BSI outcomes (higher COVID-induced diagnostic noise). The consistency across the two specifications also shows that the baseline result is not driven solely by the functional-form assumption imposed by the continuous measure of pre-pandemic hospital communication quality.

4.1.3 Pre-Trend Evidence

Figures 1 and 2 report event-study estimates based on the median-split DID specification. The omitted category is 2019.Q4, and the regressions include the same controls and fixed effects as the baseline specification. The coefficients on the interactions between the above-median communication indicator and the pre-pandemic event-time indicators provide a test for differential pre-trends.

[Insert Figure 1]

Figure 1 presents the estimates for the two BSI outcomes (higher COVID-induced diagnostic noise). For both CLABSI and MRSA, the pre-pandemic coefficients are small and statistically indistinguishable from zero, with no clear systematic trend before the onset of COVID-19. The estimates therefore show no evidence of systematic differential pre-trends. After the onset of COVID-19, the coefficients become increasingly negative, particularly in the later post-COVID periods. The pattern is especially pronounced for CLABSI and is also evident for MRSA, consistent with the negative baseline DID estimates reported in Table 5.

[Insert Figure 2]

Figure 2 reports the corresponding event-study estimates for CAUTI, SSI-CS, and SSI-AH. The pre-pandemic coefficients are again statistically indistinguishable from zero and show no consistent directional pattern. In contrast to the BSI outcomes (higher COVID-induced diagnostic noise), however, the post-COVID coefficients remain imprecisely estimated and do not display a persistent negative pattern. These results are

consistent with the baseline finding that the differential post-COVID relationship associated with higher pre-pandemic hospital communication quality is concentrated in BSI outcomes (higher COVID-induced diagnostic noise).

4.2 Alternative Explanations and Heterogeneity

The baseline results show that hospitals with higher pre-pandemic hospital communication quality experienced relatively better BSI outcomes (higher COVID-induced diagnostic noise) during COVID-19. This section examines several alternative explanations for this pattern and explores the conditions under which it is most pronounced. I first account for time-invariant hospital heterogeneity by replacing state fixed effects with hospital fixed effects. I then examine whether the baseline relationship can be explained by differences in general medical delivery quality rather than communication-related soft resilience. Next, I assess whether differences in staffing during COVID-19 account for the estimated relationships.

The analysis then turns to two dimensions of heterogeneity that speak more directly to the interpretation of the baseline findings. First, I examine heterogeneity by pre-pandemic bacterial sepsis volume. Because sepsis is closely related to bloodstream infections, greater pre-pandemic bacterial sepsis volume may proxy for greater prior BSI-related experience and more established clinical routines for recognizing and managing BSI-related cases. This test therefore examines whether the baseline relationship differs with hospitals' prior BSI-related experience. Second, I examine heterogeneity by local COVID-19 intensity. If pre-pandemic hospital communication quality serves as a proxy for communication-related soft resilience, communication-related soft resilience should become more consequential when local COVID-19 intensity is higher. I therefore test whether the relationship between pre-pandemic hospital communication quality and infection outcomes is stronger in locations and periods with greater local COVID-19 intensity.

4.2.1 Accounting for Time-Invariant Hospital Heterogeneity

One concern with the baseline specification is that pre-pandemic hospital communication quality may be correlated with persistent hospital characteristics that also affect infection outcomes. To address this concern, I replace state fixed effects with hospital fixed effects. This specification absorbs all time-invariant differences across hospitals, including persistent differences in organizational structure, patient composition, local market conditions, and other unobserved hospital characteristics. The interaction between Post and Communication_2019 remains identified because it captures whether within-hospital changes in infection rates after the onset of COVID-19 differ systematically with pre-pandemic hospital communication quality.

[Insert Table 6]

Table 6 reports the results. The estimates are nearly unchanged from the baseline specification. The coefficient on $\text{Post} \times \text{Communication_2019}$ is -0.067 for CLABSI and -0.004 for MRSA, and both remain

statistically significant at the 1 percent level. By comparison, the corresponding baseline estimates are -0.063 and -0.004. The coefficients for CAUTI, SSI-CS, and SSI-AH remain statistically indistinguishable from zero. Thus, absorbing persistent hospital-level heterogeneity leaves both the magnitude of the estimates for BSI outcomes (higher COVID-induced diagnostic noise) and the cross-outcome pattern essentially unchanged. The baseline relationship is therefore unlikely to be driven primarily by time-invariant differences between hospitals with higher and lower pre-pandemic hospital communication quality.

4.2.2 Accounting for General Medical Delivery Quality

A remaining concern is that pre-pandemic hospital communication quality may proxy for general medical delivery quality. Hospitals with higher pre-pandemic hospital communication quality may also have better clinical processes or overall quality of care, which could independently affect infection outcomes during COVID-19. Building on the hospital fixed-effects specification, I therefore examine whether the baseline results remain after allowing post-COVID infection outcomes to vary with general medical delivery quality.

I measure general medical delivery quality using the hospital-wide 30-day unplanned readmission rate. Lower readmission rates indicate better hospital performance. Readmission30_2019 is constructed from the pre-pandemic baseline data and is fixed at the hospital level before COVID-19. Communication_2019 is also fixed using the latest available pre-pandemic hospital observation from the 2018–2019 baseline sample. I interact the baseline readmission rate with the post-COVID indicator. Because both baseline measures are time invariant, their level effects are absorbed by hospital fixed effects. The interaction terms capture whether post-COVID changes in infection outcomes vary with pre-pandemic hospital communication quality and general medical delivery quality.

[Insert Table 7]

Table 7 reports the results. The coefficient on $\text{Post} \times \text{Communication}_{2019}$ is -0.070 for CLABSI and -0.004 for MRSA, and both remain statistically significant at the 1 percent level. These estimates are nearly identical to the corresponding hospital fixed-effects estimates of -0.067 and -0.004. The coefficients for CAUTI, SSI-CS, and SSI-AH remain statistically indistinguishable from zero. Thus, accounting for general medical delivery quality leaves both the magnitude of the estimates for BSI outcomes (higher COVID-induced diagnostic noise) and the cross-outcome pattern essentially unchanged.

The coefficients on $\text{Post} \times \text{Readmission30}_{2019}$ are negative across all five outcomes, although they are generally imprecisely estimated and only the estimate for SSI-AH is marginally significant. Because lower readmission rates indicate better general medical delivery quality, the signs imply, if anything, relatively larger post-COVID increases in infection rates among higher-performing hospitals. One possible explanation is patient sorting or referral selection during COVID-19, whereby higher-performing hospitals received more severely ill or clinically complex patients. Given the statistical imprecision of these estimates, however, I do

not interpret this pattern as evidence of a separate mechanism. Overall, the results suggest that the baseline relationship is unlikely to be driven by differences in general medical delivery quality.

4.2.3 Accounting for Hospital Staffing During the Pandemic

Another possible explanation is that hospitals with higher pre-pandemic hospital communication quality had greater staffing during COVID-19. Staffing is a direct input into patient care and infection prevention, and hospitals that were better able to maintain their workforce during COVID-19 may also have been better positioned to limit healthcare-associated infections (HAIs). If higher pre-pandemic hospital communication quality is systematically associated with greater staffing, the baseline estimates could partly reflect differences in labor inputs rather than communication-related soft resilience. Building on the hospital fixed-effects specification, I therefore examine whether the baseline relationship remains after accounting for hospital staffing during COVID-19.

I measure staffing using FTE_Staff, defined as the total number of full-time employees and contractors on the hospital payroll. Unlike pre-pandemic hospital communication quality and the baseline readmission measure, FTE_Staff varies over time and is observed during 2019–2022. I include both FTE_Staff and its interaction with the post-COVID indicator in the hospital fixed-effects specification. The level term accounts for within-hospital variation in staffing over time, while $\text{Post} \times \text{FTE_Staff}$ allows the relationship between staffing and infection outcomes to differ after the onset of COVID-19. This specification therefore separates the baseline relationship between pre-pandemic hospital communication quality and infection outcomes from contemporaneous differences in staffing during COVID-19.

[Insert Table 8]

Table 8 reports the results. Accounting for hospital staffing leaves the estimates on pre-pandemic hospital communication quality essentially unchanged. The coefficient on $\text{Post} \times \text{Communication_2019}$ is -0.067 for CLABSI and -0.004 for MRSA, and both remain statistically significant at the 1 percent level. These estimates are identical in magnitude to the corresponding hospital fixed-effects estimates reported in Table 6. The coefficients for CAUTI, SSI-CS, and SSI-AH remain statistically indistinguishable from zero. Thus, controlling for time-varying staffing preserves both the magnitude of the estimates for BSI outcomes (higher COVID-induced diagnostic noise) and the distinction between BSI outcomes (higher COVID-induced diagnostic noise) and non-BSI outcomes (lower COVID-induced diagnostic noise).

The coefficients on $\text{Post} \times \text{FTE_Staff}$ are statistically insignificant across all five infection outcomes and do not exhibit a consistent directional pattern. This provides little indication that staffing during COVID-19 accounts for the association between pre-pandemic hospital communication quality and BSI outcomes (higher COVID-induced diagnostic noise). Overall, the baseline pattern remains essentially unchanged after controlling for both within-hospital changes in staffing and differential post-COVID relationships between

staffing and infection rates.

4.2.4 Heterogeneity by Pre-Pandemic Bacterial Sepsis Volume

I next examine whether the relationship between pre-pandemic hospital communication quality and infection outcomes varies with hospitals’ prior BSI-related experience. Bloodstream infection and bacterial sepsis are distinct clinical conditions, but they are closely related. Bloodstream infections can lead to bacterial sepsis, and both conditions can initially present with nonspecific systemic symptoms rather than clearly localized signs (Huerta and Rice, 2019). Their early evaluation and management also overlap in important respects, including recognition of systemic deterioration, blood-culture collection, antimicrobial treatment, and escalation of care.

This recognition problem became more difficult during COVID-19. Severe COVID-19 can itself produce viral sepsis, and clinical evidence shows that its manifestations overlap with those of bacterial sepsis (Walsh et al., 2023). Walsh et al. (2023) describe the convergence of severe COVID-19 viral sepsis and bacterial sepsis as posing a “daunting bedside challenge to accurate diagnosis”. They further emphasize that existing diagnostic standards make it difficult to distinguish critically ill patients with COVID-19 viral sepsis from patients with bacterial sepsis. Thus, although BSI and bacterial sepsis are not identical conditions, they share an important early-recognition problem, and clinical evidence indicates that COVID-19 increased the diagnostic noise surrounding this problem. Pre-pandemic experience with bacterial sepsis therefore provides a useful proxy for hospitals’ prior exposure to clinical situations and response processes that are also relevant to the recognition and management of bloodstream infections.

Sepsis management also depends on structured clinical routines. SEP-1 is a process measure centered on the timely recognition and treatment of severe sepsis and septic shock. Its components include blood-culture collection, antibiotic administration, lactate measurement, fluid resuscitation, reassessment, and other time-sensitive interventions. (Warstadt et al., 2022) show that sepsis bundle performance depends on coordinated processes involving EHR-based alerts, provider and nurse education, bedside huddles, checklists, and repeated performance feedback (Warstadt et al., 2022). They also emphasize that the bundle extends across multiple hours and transitions of care, making later components particularly vulnerable to errors or omissions and increasing the value of system-wide processes (Warstadt et al., 2022). Consistent with the strain imposed by COVID-19, the study reports declining sepsis bundle compliance during the height of the pandemic. These findings support the interpretation that repeated exposure to sepsis cases provides hospitals with opportunities to develop and reinforce recognition-and-response routines that may also be relevant for BSI management.

I construct a proxy for pre-pandemic bacterial sepsis volume using hospitals’ SEP-1 reporting status. CMS reporting requirements make SEP-1 availability informative about underlying case volume. Hospitals with

five or fewer eligible sepsis cases in a quarter are not required to submit patient-level SEP-1 data ⁵, while very small denominators may also be suppressed from public reporting ⁶. Consistent with these rules, prior evidence shows that low case volume is a major reason for SEP-1 nonreporting and that reporting hospitals tend to have greater bacterial sepsis volume and more developed sepsis quality-improvement structures (Bolte et al., 2022). These differences make SEP-1 reporting status informative about hospitals' prior exposure to severe sepsis cases and the clinical routines associated with their management.

Low_Volume is constructed from SEP-1 reporting in the 2018–2019 pre-pandemic baseline sample. A hospital is classified as Low_Volume if it has at least one missing SEP-1 report during this period. Hospitals without a missing SEP-1 report during the baseline period form the comparison group. Low_Volume is intended to proxy for lower pre-pandemic bacterial sepsis volume and, in turn, less prior BSI-related experience. It does not directly measure bacterial sepsis volume.

Hospitals with greater pre-pandemic bacterial sepsis volume may have greater prior BSI-related experience before COVID-19. This prior BSI-related experience may reduce the additional noise introduced by COVID-19 into BSI-related tasks. In contrast, Low_Volume hospitals may have less prior BSI-related experience, so the same COVID-19 shock may generate greater effective task-related noise in recognizing and managing bacterial sepsis and BSI. Communication-related soft resilience should therefore become more consequential for post-COVID BSI outcomes (higher COVID-induced diagnostic noise) among Low_Volume hospitals, where effective task-related noise in BSI recognition and management may be greater. I test this prediction using a triple-interaction specification among Post, Communication_2019, and Low_Volume, with hospital and year-quarter fixed effects. The coefficient on $\text{Post} \times \text{Communication}_{2019} \times \text{Low_Volume}$ measures whether the post-COVID relationship between pre-pandemic hospital communication quality and BSI outcomes (higher COVID-induced diagnostic noise) is stronger among Low_Volume hospitals than among hospitals not classified as Low_Volume.

[Insert Table 9]

Table 9 reports the results. The clearest evidence of heterogeneity appears for MRSA. The coefficient on $\text{Post} \times \text{Communication}_{2019}$ is -0.001 and statistically insignificant. The coefficient on $\text{Post} \times \text{Communication}_{2019} \times \text{Low_Volume}$ is -0.008 and statistically significant at the 5 percent level. The implied $\text{Post} \times \text{Communication}_{2019}$ coefficient for Low_Volume hospitals is therefore approximately -0.009. Thus, the negative post-COVID relationship between pre-pandemic hospital communication quality and MRSA is substantially stronger among Low_Volume hospitals. This pattern is consistent with the effective task-related noise interpretation. Low_Volume hospitals may have less prior BSI-related experience and therefore

⁵https://www.qualityreportingcenter.com/globalassets/iqr2019events/iqr05292019/iqr_qa-summary_fy2021chrtabstval_05292019_vfinal508.pdf

⁶https://www.qualityreportingcenter.com/globalassets/iqr_resources/november-2019/pr_jan-2020_ip-hc-preview_help-guide_vfinal508.pdf

face greater effective task-related noise in BSI recognition and management during COVID-19, making communication-related soft resilience more consequential for their BSI outcomes (higher COVID-induced diagnostic noise).

For CLABSI, the evidence points in the same direction but is much weaker. The triple-interaction coefficient is negative (-0.005), but it is not statistically distinguishable from zero. The coefficient on $\text{Post} \times \text{Communication}_{2019}$ is -0.045 and statistically significant at the 10 percent level. The implied coefficient for Low_Volume hospitals is therefore approximately -0.050. The negative triple interaction is directionally consistent with the prediction that communication-related soft resilience should become more consequential when effective task-related noise in BSI recognition and management is greater. However, the difference between Low_Volume hospitals and hospitals not classified as Low_Volume is small and imprecisely estimated. The CLABSI results therefore provide only limited support for this heterogeneity.

The non-BSI outcomes (lower COVID-induced diagnostic noise) do not show a comparable pattern. The triple-interaction coefficients for CAUTI and SSI-CS are negative but statistically insignificant. For SSI-AH, the coefficient is positive (0.276) and statistically significant at the 10 percent level. This estimate implies that the post-COVID relationship between pre-pandemic hospital communication quality and infection outcomes is less negative, or more positive, among Low_Volume hospitals for SSI-AH. Its direction is therefore opposite to the effective task-related noise interpretation. However, this pattern appears only for SSI-AH. Neither CAUTI nor SSI-CS shows statistically significant heterogeneity by Low_Volume. The SSI-AH estimate therefore does not indicate a broader pattern across non-BSI outcomes (lower COVID-induced diagnostic noise). Given its marginal significance, its opposite sign, and the number of outcomes examined, I interpret it cautiously as possible outcome-specific variation or statistical noise rather than evidence of a systematic non-BSI relationship.

The lower-order interaction between Low_Volume and Post is negative for four of the five outcomes, although the pattern is not uniform and the MRSA coefficient is positive. These estimates may reflect pandemic-related changes in patient composition rather than the proposed communication-related soft resilience interpretation. Hospitals without reported SEP-1 measures are disproportionately represented among smaller and critical-access hospitals (Bolte et al., 2022). During COVID-19, higher-risk or more complex patients may have delayed care, bypassed smaller hospitals, or been referred to larger centers. Such changes could affect measured infection rates independently of pre-pandemic hospital communication quality. Because the lower-order coefficients vary across outcomes, I treat this possibility as suggestive rather than as a separate finding.

Low_Volume is also an imperfect proxy for prior BSI-related experience. Missing SEP-1 performance is informative about low bacterial sepsis volume under CMS reporting rules, but nonreporting does not always imply low bacterial sepsis volume. Missing values may also arise for other reporting or data-related reasons. In addition, bacterial sepsis volume is not identical to BSI volume. Bacterial sepsis volume therefore does

not map one-for-one into hospitals' prior exposure to BSI cases. Low_Volume should instead be interpreted as a proxy for prior BSI-related experience. Measurement error in this proxy may reduce the precision of the estimated heterogeneity.

Overall, the results provide suggestive and outcome-specific evidence for the effective task-related noise interpretation. For MRSA, the negative post-COVID relationship between pre-pandemic hospital communication quality and BSI outcomes (higher COVID-induced diagnostic noise) is substantially stronger among Low_Volume hospitals. This pattern is consistent with Low_Volume hospitals having less prior BSI-related experience and facing greater effective task-related noise in BSI recognition and management during COVID-19. For CLABSI, the triple-interaction estimate has the predicted negative sign but is small and imprecisely estimated. The non-BSI outcomes (lower COVID-induced diagnostic noise) do not show a consistent pattern: CAUTI and SSI-CS exhibit no statistically significant heterogeneity, while SSI-AH produces a marginally significant estimate in the opposite direction. Taken together, the evidence is most consistent with the effective task-related noise interpretation for MRSA, with weaker support for CLABSI and no systematic evidence across the non-BSI outcomes (lower COVID-induced diagnostic noise).

4.2.5 Heterogeneity by Local COVID-19 Intensity

I next examine whether the relationship between pre-pandemic hospital communication quality and infection outcomes becomes stronger when local COVID-19 intensity is higher. If pre-pandemic hospital communication quality serves as a proxy for communication-related soft resilience, communication-related soft resilience should become more consequential when local COVID-19 intensity is higher. This interpretation yields a natural heterogeneity prediction: hospitals with higher pre-pandemic hospital communication quality should experience relatively better infection outcomes when local COVID-19 intensity is higher.

Because hospital-level operational strain is not directly observed, I construct a county-quarter local COVID-19 intensity index using new COVID-19 cases and deaths. The relationship between local disease burden and hospital stress need not be linear. Once a hospital is already operating under substantial pandemic pressure, further increases in raw case or death counts may not translate proportionally into additional effective strain. Using raw counts would also place substantial weight on extreme observations. I therefore convert county-level new cases and deaths into percentile ranks and rescale each component to the $[0, 1]$ interval. Covid_Intensity is defined as the equally weighted average of the two percentile ranks. This rank-based transformation places cases and deaths on a common scale, reduces the influence of extreme observations, and allows the measure to capture the relative severity of the local pandemic environment without imposing a linear relationship between raw epidemiological counts and hospital stress.

Formally, for each county-quarter observation,

$$P(X_{ct}) = \frac{r(X_{ct}) - 1}{N_X - 1},$$

where X_{ct} denotes either county-level new COVID-19 cases or deaths. The rank $r(X_{ct})$ is calculated using all non-missing observations for that variable in the full estimation sample, pooling observations across all counties and time periods. The ranking is therefore not constructed separately within quarters, years, counties, or other subgroups. N_X denotes the total number of non-missing observations for the corresponding component. I then define

$$CovidIntensity_{ct} = \frac{P(Cases_{ct}) + P(Deaths_{ct})}{2}.$$

Higher values therefore indicate county-quarter observations that rank higher in the overall sample distribution of both COVID-19 cases and deaths.

I interact Covid.Intensity with the median-split measure of pre-pandemic hospital communication quality in the hospital fixed-effects specification. Communication_2019_median equals one for hospitals with above-median pre-pandemic hospital communication quality. The median split is based on the same pre-pandemic hospital communication quality measure used in the baseline analysis. Hospital fixed effects absorb persistent differences across hospitals, while year-quarter fixed effects absorb shocks common to all hospitals in a given period. The interaction Covid.Intensity $\times$ Communication_2019_median therefore captures whether the relationship between local COVID-19 intensity and infection rates differs between hospitals with above- and at-or-below-median pre-pandemic hospital communication quality.

[Insert Table 10]

Table 10 reports the results. The interaction between Covid.Intensity and Communication_2019_median is negative for both BSI outcomes (higher COVID-induced diagnostic noise). For CLABSI, the interaction coefficient is -0.120 and statistically significant at the 10 percent level. For MRSA, the coefficient is -0.009 and statistically significant at the 5 percent level. These estimates indicate that the relationship between local COVID-19 intensity and bloodstream infection rates differs systematically with pre-pandemic hospital communication quality. As local COVID-19 intensity increases, infection rates change relatively less at hospitals with above-median pre-pandemic hospital communication quality than at hospitals at or below the median.

The lower-order Covid.Intensity coefficient provides additional context. For CLABSI, Covid.Intensity is positively associated with infection rates among hospitals with at-or-below-median pre-pandemic hospital communication quality, with a coefficient of 0.275 that is statistically significant at the 1 percent level. The negative interaction implies that this positive relationship is substantially attenuated among hospitals

with above-median pre-pandemic hospital communication quality. The corresponding implied slope for hospitals with above-median pre-pandemic hospital communication quality is approximately 0.155. For MRSA, the Covid_Intensity coefficient is small and statistically insignificant, but the negative and significant interaction indicates that the relationship between local COVID-19 intensity and MRSA rates nevertheless differs systematically across the above- and at-or-below-median pre-pandemic hospital communication quality groups.

The non-BSI outcomes (lower COVID-induced diagnostic noise) do not display a comparable pattern. The interaction coefficients for CAUTI, SSI-CS, and SSI-AH are all statistically indistinguishable from zero and do not exhibit a common directional response. Thus, the differential relationship between pre-pandemic hospital communication quality and local COVID-19 intensity is again concentrated in BSI outcomes (higher COVID-induced diagnostic noise) rather than non-BSI outcomes (lower COVID-induced diagnostic noise).

Overall, the results are consistent with the communication-related soft resilience interpretation of the baseline findings. The relationship between pre-pandemic hospital communication quality and BSI outcomes (higher COVID-induced diagnostic noise) becomes more pronounced when local COVID-19 intensity is higher, while no comparable pattern appears for the non-BSI outcomes (lower COVID-induced diagnostic noise). This pattern is consistent with communication-related soft resilience becoming more consequential when local COVID-19 intensity is higher.

4.3 Robustness

This section examines whether the baseline findings are robust to alternative interpretations, measurement choices, and sample composition. I first assess the discriminant validity of the communication measure by examining whether the results can be distinguished from other dimensions of hospital quality that are less closely related to communication-related soft resilience. I then consider alternative communication measures to assess whether the findings depend on the particular construction of the baseline communication measure. Finally, I re-estimate the baseline specification across an alternative subsample to examine whether the results are driven by hospitals with intermediate communication scores.

4.3.1 Discriminant Validity of the Communication Measure

A potential concern with the baseline results is that pre-pandemic hospital communication quality may capture general patient experience rather than communication-related soft resilience specifically. To assess this possibility, I use pre-pandemic hospital quietness as a discriminant measure. Quietness provides a useful comparison because it is measured within the same HCAHPS patient-experience framework as communication but captures a conceptually distinct dimension of the hospital experience.

Existing psychometric evidence supports this distinction. [Arah et al. \(2006\)](#) identify doctor communication,

nurse communication, and the physical hospital environment as distinct dimensions of patient experience. In their factor analysis, the item measuring whether the hospital room was quiet at night loads on the physical-environment factor rather than on either communication factor. Consistent with this interpretation, [Bliefnick et al. \(2019\)](#) relate HCAHPS quietness ratings to objectively measured acoustic conditions in hospital rooms and find that several acoustic measures are significantly associated with patients’ reported quietness. Quietness therefore provides a useful discriminant measure: it shares the patient-reported HCAHPS measurement framework with communication while capturing a dimension of hospital experience more closely related to the physical and acoustic environment.

This comparison helps distinguish the communication-related soft resilience interpretation from a general patient-experience explanation. If the baseline pattern for BSI outcomes (higher COVID-induced diagnostic noise) simply reflected a general tendency for hospitals with better pre-pandemic patient-experience ratings to perform better during COVID-19, a similar negative relationship should appear for quietness. By contrast, if the results are specific to communication-related soft resilience, the $\text{Post} \times \text{Communication}_{2019}$ coefficients should remain after accounting for quietness, while $\text{Post} \times \text{Quietness}$ need not reproduce the same pattern.

I therefore augment the baseline specification with pre-pandemic quietness and its interaction with the post-COVID indicator. Quietness and $\text{Communication}_{2019}$ are constructed from the same baseline observation for each hospital. Specifically, both use the latest available pre-pandemic HCAHPS observation from 2018–2019. This common baseline observation ensures that their comparison does not reflect differences in measurement timing within hospitals. The specification includes both $\text{Post} \times \text{Communication}_{2019}$ and $\text{Post} \times \text{Quietness}$. It therefore distinguishes differential post-COVID changes associated with pre-pandemic hospital communication quality from those associated with pre-pandemic hospital quietness.

[Insert Table 11]

Table 11 reports the results. The coefficients on $\text{Post} \times \text{Communication}_{2019}$ remain negative and statistically significant for both BSI outcomes (higher COVID-induced diagnostic noise). The coefficient is -0.084 for CLABSI and -0.006 for MRSA, both significant at the 1 percent level. These estimates are somewhat larger in magnitude than the corresponding baseline estimates of -0.063 and -0.004. Thus, accounting for pre-pandemic quietness does not attenuate the relationship between pre-pandemic hospital communication quality and post-COVID changes in BSI outcomes.

In contrast, quietness does not reproduce the communication pattern. The coefficient on $\text{Post} \times \text{Quietness}$ is 0.025 for CLABSI and statistically indistinguishable from zero. For MRSA, the coefficient is 0.003 and statistically significant at the 5 percent level, but its sign is positive, opposite to the negative coefficient on communication. The quietness interactions for CAUTI, SSI-CS, and SSI-AH are also statistically insignificant. There is therefore no corresponding pattern in which greater pre-pandemic quietness is associated

with smaller post-COVID changes in BSI rates.

Overall, the results provide evidence consistent with the discriminant validity of the communication measure. Another patient-reported HCAHPS measure drawn from the same measurement framework but capturing a distinct dimension of hospital experience does not reproduce the concentrated negative pattern for BSI outcomes observed for communication. At the same time, controlling for quietness leaves the communication coefficients for CLABSI and MRSA intact. The baseline findings are therefore unlikely to reflect a general patient-experience component common to HCAHPS measures alone.

4.3.2 Alternative Communication Measures

A second concern is that the baseline findings may depend on the particular construction of the communication measure. To examine this possibility, I replace Communication_2019 with four alternative communication measures: Nurse Communication, Doctor Communication, Staff Responsiveness, and Communication about Medicines. These measures capture related but distinct dimensions of information exchange within the hospital and provide alternative proxies for communication-related soft resilience.

All four alternative measures are constructed from the same baseline observation as Communication_2019. For each hospital, I use the latest available pre-pandemic HCAHPS observation from 2018–2019. Each measure is reported on the same 1–5 scale. Holding the baseline observation and scale fixed ensures that differences across specifications reflect the alternative communication measures rather than differences in measurement timing or normalization. I re-estimate the baseline DID specification separately using each alternative measure and its interaction with the post-COVID indicator.

[Insert Table 12]

Table 12 first reports the results using Nurse Communication. The coefficient on $\text{Post} \times \text{Nurse Communication}$ is -0.052 for CLABSI and -0.004 for MRSA, both statistically significant at the 1 percent level. Relative to the corresponding sample means, these estimates amount to approximately 8.4 percent of the mean CLABSI rate and 10.0 percent of the mean MRSA rate for a one-point increase in Nurse Communication. In contrast, the interaction coefficients for CAUTI, SSI-CS, and SSI-AH are statistically indistinguishable from zero. Thus, replacing the baseline measure with Nurse Communication reproduces the same concentration of negative post-COVID relationships in the two BSI outcomes (higher COVID-induced diagnostic noise).

[Insert Table 13]

Table 13 uses Doctor Communication. The coefficient on $\text{Post} \times \text{Doctor Communication}$ is -0.047 for CLABSI and statistically significant at the 1 percent level, corresponding to approximately 7.6 percent of the sample mean. The MRSA coefficient is also negative (-0.002), equivalent to approximately 5.0 percent of its sample mean, although it is not statistically distinguishable from zero. None of the three interaction

coefficients for non-BSI outcomes (lower COVID-induced diagnostic noise) is statistically significant. The same cross-outcome pattern therefore remains, although the MRSA estimate is less precisely estimated when communication is measured using Doctor Communication.

[Insert Table 14]

Table 14 replaces the baseline measure with Staff Responsiveness. The coefficient on $\text{Post} \times \text{Staff Responsiveness}$ is -0.047 for CLABSI and -0.003 for MRSA, statistically significant at the 5 and 1 percent levels, respectively. Relative to the corresponding sample means, these estimates amount to approximately 7.6 percent of the mean CLABSI rate and 7.5 percent of the mean MRSA rate for a one-point increase in Staff Responsiveness. The interaction coefficients for CAUTI, SSI-CS, and SSI-AH remain statistically insignificant. The pattern for BSI outcomes (higher COVID-induced diagnostic noise) therefore again appears when Staff Responsiveness is used as the alternative communication measure.

[Insert Table 15]

Finally, Table 15 uses Communication about Medicines. The coefficient on $\text{Post} \times \text{Communication about Medicines}$ is -0.047 for CLABSI and -0.003 for MRSA, statistically significant at the 1 and 5 percent levels, respectively. These estimates correspond to approximately 7.6 percent of the mean CLABSI rate and 7.5 percent of the mean MRSA rate for a one-point increase in Communication about Medicines. As in the other specifications, none of the interaction coefficients for the three non-BSI outcomes (lower COVID-induced diagnostic noise) is statistically distinguishable from zero. Communication about Medicines therefore produces the same qualitative pattern for BSI outcomes observed with the baseline measure.

Taken together, the alternative communication measures produce a highly consistent pattern. All four specifications yield negative and statistically significant interaction coefficients for CLABSI. All four MRSA coefficients are also negative, with three statistically significant and the Doctor Communication estimate imprecisely estimated. In contrast, none of the twelve corresponding interaction coefficients across CAUTI, SSI-CS, and SSI-AH is statistically significant. The estimated magnitudes for the BSI outcomes are also economically meaningful, generally corresponding to approximately 5–10 percent of the respective sample means for a one-point increase in the alternative communication measures. The baseline findings therefore do not depend on the particular construction of $\text{Communication}_{2019}$. The concentration of the post-COVID relationship in BSI outcomes appears across multiple dimensions of pre-pandemic hospital communication measured using the same baseline observation.

4.3.3 Subsample Robustness

A final concern is that the baseline findings may be sensitive to hospitals with intermediate communication scores. Hospitals near the center of the communication distribution differ only modestly in measured commu-

nication quality, so small differences in survey responses or measurement error may obscure the underlying contrast in communication-related soft resilience. This issue is particularly relevant for specifications that compare hospitals across different levels of pre-pandemic hospital communication quality.

I therefore construct a more restrictive subsample that excludes hospitals with `Communication_2019` scores from 2.8 through 3.2. The restricted sample retains only hospitals with `Communication_2019` greater than 3.2 or below 2.8. This focuses the analysis on hospitals with more clearly separated levels of pre-pandemic hospital communication quality. `Communication_2019` continues to use each hospital’s latest available pre-pandemic HCAHPS observation from 2018–2019. This restriction changes only the composition of the estimation sample; the outcome definitions, control variables, fixed effects, and clustering structure remain unchanged.

I re-estimate the baseline DID specification in this restricted sample using the continuous communication measure. This exercise tests whether the baseline pattern persists after removing hospitals with relatively similar communication scores in the middle of the distribution.

[Insert Table 16]

Table 16 reports the results. The coefficient on $\text{Post} \times \text{Communication_2019}$ is -0.063 for CLABSI and is statistically significant at the 1 percent level. For MRSA, the corresponding coefficient is -0.004 and statistically significant at the 5 percent level. These estimates are essentially identical to the baseline continuous-DID estimates of -0.063 and -0.004, respectively. Thus, removing hospitals with intermediate communication scores leaves both the magnitude and statistical significance of the relationships for BSI outcomes (higher COVID-induced diagnostic noise) unchanged.

The non-BSI outcomes (lower COVID-induced diagnostic noise) also reproduce the baseline pattern. The interaction coefficients are -0.006 for CAUTI, 0.025 for SSI-CS, and 0.040 for SSI-AH, and all three are statistically indistinguishable from zero. The negative post-COVID relationship therefore remains concentrated in CLABSI and MRSA rather than extending to non-BSI outcomes.

Overall, the subsample results indicate that the baseline findings are not driven by hospitals with communication scores clustered in the middle of the distribution. Even after restricting the sample to hospitals with more clearly differentiated levels of pre-pandemic hospital communication quality, the estimated coefficients for BSI outcomes remain virtually unchanged, while the non-BSI outcomes continue to show no systematic relationship. The main pattern is therefore robust to this alternative sample composition.

5 Interpretation and Limitations

The results point to a common pattern. In this study, pre-pandemic hospital communication quality serves as a proxy for communication-related soft resilience. Its relationship with infection-control performance becomes stronger when effective task-related noise is greater. The evidence for this interpretation comes from three dimensions of heterogeneity: COVID-induced diagnostic noise across infection-control tasks, prior BSI-related experience, and local COVID-19 intensity.

The first dimension is COVID-induced diagnostic noise across infection-control tasks. The baseline relationship is concentrated in BSI outcomes (higher COVID-induced diagnostic noise). Higher pre-pandemic hospital communication quality is associated with smaller post-COVID changes in CLABSI and MRSA rates, while no comparable pattern appears for CAUTI, SSI-CS, or SSI-AH. This distinction is consistent with the proposed interpretation. COVID-19 increased the diagnostic noise surrounding BSI-related tasks, making communication-related soft resilience more consequential for these tasks. The absence of a similar pattern across the non-BSI outcomes (lower COVID-induced diagnostic noise) also suggests that pre-pandemic hospital communication quality does not simply predict lower infection rates in general.

The second dimension is prior BSI-related experience. Hospitals with greater pre-pandemic bacterial sepsis volume may have had greater prior BSI-related experience. Such experience may have reduced the additional noise created by COVID-19. Hospitals classified as Low_Volume are more likely to have had less prior BSI-related experience, so the same pandemic shock may have generated greater effective task-related noise. The results are consistent with this prediction. For MRSA, the negative post-COVID relationship between pre-pandemic hospital communication quality and infection outcomes is substantially stronger among Low_Volume hospitals. The CLABSI estimate has the same sign, although the difference is small and imprecisely estimated. The evidence is therefore stronger for MRSA than for CLABSI. More broadly, the pattern is consistent with communication-related soft resilience becoming more consequential when prior BSI-related experience provides less protection against greater effective task-related noise in BSI recognition and management.

The third dimension is local COVID-19 intensity. Higher local COVID-19 intensity exposed hospitals to greater pandemic-related disruption. The relationship between pre-pandemic hospital communication quality and BSI outcomes (higher COVID-induced diagnostic noise) becomes stronger as local COVID-19 intensity increases. This pattern appears for both CLABSI and MRSA. No comparable pattern appears for the non-BSI outcomes (lower COVID-induced diagnostic noise). The local COVID-19 intensity results therefore provide a separate source of variation in the conditions under which communication-related soft resilience becomes more consequential.

Taken together, these results provide a common interpretation of the empirical findings. Pre-pandemic hospi-

tal communication quality does not predict uniformly better outcomes across tasks, hospitals, and pandemic conditions. Its relationship with infection outcomes is concentrated in BSI outcomes (higher COVID-induced diagnostic noise), stronger when hospitals have less prior BSI-related experience, and stronger when local COVID-19 intensity is higher. This pattern is consistent with pre-pandemic hospital communication quality serving as a proxy for communication-related soft resilience and with communication-related soft resilience becoming more consequential as effective task-related noise increases.

The full set of results is also difficult to reconcile with several simpler explanations. One possibility is that pre-pandemic hospital communication quality captures persistent differences across hospitals. The estimates for BSI outcomes (higher COVID-induced diagnostic noise), however, remain nearly unchanged after replacing state fixed effects with hospital fixed effects. A second possibility is that pre-pandemic hospital communication quality proxies for general medical delivery quality. Accounting for pre-pandemic readmission performance leaves the estimates on pre-pandemic hospital communication quality largely unchanged. A third possibility is that hospitals with higher pre-pandemic hospital communication quality maintained greater staffing during COVID-19. Controlling for time-varying staffing also leaves the main pattern for BSI outcomes essentially unchanged. The results are similarly difficult to explain by the general experience of the patient. Hospital quietness is measured within the same HCAHPS framework as communication but does not reproduce the same negative pattern for BSI outcomes. More importantly, a general explanation of patient experience does not naturally predict the three dimensions of heterogeneity documented above. It does not explain why the relationship is concentrated in BSI outcomes, stronger among hospitals with less prior BSI-related experience, and stronger when local COVID-19 intensity is higher.

Changes in patient composition remain a possible concern, although the analysis addresses one important form of this explanation. During COVID-19, higher-risk or more complex patients may have been more likely to seek care in, or be referred to, hospitals with better pre-pandemic medical performance. If this type of sorting were an important driver of the results, controlling for differential post-COVID changes by pre-pandemic medical delivery quality should reduce the estimated communication relationship. However, adding the interaction between the post-COVID period and the pre-pandemic 30-day readmission rate leaves the estimates for BSI outcomes essentially unchanged. This makes sorting based on baseline medical quality less consistent with the results. The composition of the patients may still have changed through other channels, especially in smaller hospitals or Low_Volume, so patient sorting cannot be completely ruled out. It is also unclear why sorting alone would produce the full pattern in COVID-induced diagnostic noise, prior experience related to BSI, and local COVID-19 intensity. Therefore, changes in patient composition are considered a remaining alternative explanation rather than a complete explanation for the results.

Several limitations qualify the interpretation. First, pre-pandemic hospital communication quality is not randomly assigned. COVID-19 provides an external shock, and pre-pandemic hospital communication qual-

ity is measured before that shock. Hospitals with different pre-pandemic hospital communication scores may nevertheless differ in ways that affect their responses to COVID-19. Hospital fixed effects and the additional controls address several observable and time-invariant differences, but they cannot eliminate all possible time-varying confounding. The estimates should therefore be interpreted as differential post-COVID relationships associated with pre-pandemic hospital communication quality rather than as the causal effect of communication itself.

Second, effective task-related noise is not directly observed. The interpretation relies instead on three dimensions along which effective task-related noise should differ: COVID-induced diagnostic noise across infection-control tasks, prior BSI-related experience, and local COVID-19 intensity. The relationship between pre-pandemic hospital communication quality and infection outcomes becomes stronger along these dimensions, which supports the proposed effective task-related noise interpretation. However, these heterogeneity patterns do not uniquely identify the underlying mechanism. Other factors correlated with these dimensions may also affect the relationship between communication-related soft resilience and infection-control performance. The results therefore support the proposed interpretation but cannot rule out all alternative explanations.

The measures used in the heterogeneity analysis are also imperfect. The distinction between BSI outcomes (higher COVID-induced diagnostic noise) and non-BSI outcomes (lower COVID-induced diagnostic noise) does not capture every difference in diagnostic noise. SEP-1 reporting status is an imperfect proxy for pre-pandemic bacterial sepsis volume and prior BSI-related experience. In the empirical construction, Low_Volume identifies hospitals with at least one missing SEP-1 report in the 2018–2019 pre-pandemic baseline sample. Missing SEP-1 reporting is informative about low bacterial sepsis volume under CMS reporting rules, but it does not uniquely identify low bacterial sepsis volume. Bacterial sepsis volume is also not identical to BSI volume. Local COVID-19 intensity does not directly measure the conditions faced by each hospital. These measurement limitations may weaken the estimated heterogeneity and limit how precisely each source of effective task-related noise can be identified.

The heterogeneity evidence is also not uniform across outcomes. The Low_Volume result is pronounced for MRSA but much weaker for CLABSI. In the same analysis, SSI-AH produces a marginally significant estimate in the opposite direction. No single heterogeneity result therefore provides conclusive evidence for the proposed interpretation. The interpretation instead rests on the common pattern across the three dimensions. Each points toward a stronger role for communication-related soft resilience when effective task-related noise is expected to be greater, but none rules out other mechanisms operating at the same time.

Finally, the results come from an unusually severe external shock. COVID-19 created substantial disruption and increased effective task-related noise. The findings therefore do not imply that communication-related

soft resilience is equally consequential during normal periods. Rather, its relationship with infection-control performance becomes stronger when effective task-related noise is greater. Pre-pandemic hospital communication quality, as its proxy, is most strongly related to infection-control performance under greater effective task-related noise and disruption.

6 Conclusion

This paper studies when communication-related soft resilience becomes more consequential for infection-control performance. I use pre-pandemic hospital communication quality as a proxy for this capacity and the onset of COVID-19 as a common external shock. The empirical analysis asks whether infection outcomes changed differently after COVID-19 across hospitals with different levels of pre-pandemic hospital communication quality.

The results show a clear pattern. The relationship between pre-pandemic hospital communication quality and post-COVID infection outcomes is concentrated in BSI outcomes with higher COVID-induced diagnostic noise. No comparable pattern appears for non-BSI outcomes with lower COVID-induced diagnostic noise. The same relationship becomes stronger along two additional margins of effective task-related noise. It is stronger among hospitals with less prior BSI-related experience and as local COVID-19 intensity increases. These patterns remain after accounting for time-invariant hospital heterogeneity, pre-pandemic medical delivery quality, staffing, general patient experience, alternative communication measures, and alternative sample composition.

Taken together, the findings suggest that communication-related soft resilience becomes more consequential as effective task-related noise increases. Pre-pandemic hospital communication quality does not predict uniformly better infection outcomes. Its relationship with infection-control performance is stronger when effective task-related noise is greater. This pattern is consistent with a broader organizational argument: physical resources alone do not determine performance under stress, and communication-related soft resilience can become more consequential when organizations face greater task-related noise.

The evidence should not be interpreted as the causal effect of communication itself. Pre-pandemic hospital communication quality is not randomly assigned, and effective task-related noise is not directly observed. The findings instead document a common pattern across infection-control tasks, prior BSI-related experience, and local COVID-19 intensity. Across these settings, communication-related soft resilience is most strongly associated with infection-control performance when effective task-related noise is greater.

References

Alp, Emine, Hakan Leblebicioglu, Mehmet Doganay, and Andreas Voss, “Infection control practice

- in countries with limited resources,” *Annals of clinical microbiology and antimicrobials*, 2011, 10, 1–4.
- Arah, Onyebuchi A, AHA Ten Asbroek, Diana MJ Delnoij, Johan S De Koning, Piet JA Stam, Aldien H Poll, Barbara Vriens, Paul F Schmidt, and Niek S Klazinga**, “Psychometric Properties of the Dutch Version of the Hospital-Level Consumer Assessment of Health Plans Survey® Instrument,” *Health services research*, 2006, 41 (1), 284–301.
- Aslam, Saif, Habtamu Asrat, Rina Liang, Wenqi Qiu, Subin Sunny, Anna Maro, Marie Abdallah, Mary Fornek, Briana Episcopia, and John Quale**, “Methicillin-resistant *Staphylococcus aureus* bacteremia during the coronavirus disease 2019 (COVID-19) pandemic: Trends and distinguishing characteristics among patients in a healthcare system in New York City,” *Infection Control & Hospital Epidemiology*, 2023, 44 (7), 1177–1179.
- Barowskin, Janelle**, “Increases in Hospital-Acquired COVID-19 Linked to High Community-Onset Incidence.,” *Neurology Advisor*, 2023, pp. NA–NA.
- Battiston, Diego, Jordi Blanes i Vidal, and Tom Kirchmaier**, “Face-to-face communication in organizations,” *The Review of Economic Studies*, 2021, 88 (2), 574–609.
- Belasen, Alan T, Jane Oppenlander, Ariel R Belasen, and Attila J Hertelendy**, “Provider–patient communication and hospital ratings: perceived gaps and forward thinking about the effects of COVID-19,” *International Journal for Quality in Health Care*, 2021, 33 (1), mzaa140.
- Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan**, “How much should we trust differences-in-differences estimates?,” *The Quarterly journal of economics*, 2004, 119 (1), 249–275.
- Bliefnick, Jay M, Erica E Ryherd, and Rebecca Jackson**, “Evaluating hospital soundscapes to improve patient experience,” *The Journal of the Acoustical Society of America*, 2019, 145 (2), 1117–1128.
- Bloom, Nicholas, Carol Propper, Stephan Seiler, and John Van Reenen**, “The impact of competition on management quality: evidence from public hospitals,” *The Review of Economic Studies*, 2015, 82 (2), 457–489.
- Bogusz, Alexandra, Munro Stewart, Jennifer Hunter, Brigitte Yip, Damien Reid, Chris Robertson, and Stephanie J Dancer**, “How quickly do hospital surfaces become contaminated after detergent cleaning?,” *Healthcare infection*, 2013, 18 (1), 3–9.
- Bolte, Ty B, Morgan B Swanson, Anna M Kaldjian, Nicholas M Mohr, Jennifer McDanel, and Azeemuddin Ahmed**, “Hospitals that report severe sepsis and septic shock bundle compliance have more structured sepsis performance improvement,” *Journal of patient safety*, 2022, 18 (8), e1231–e1236.

- Bono, S De, G Heling, and Michael Angelo Borg**, “Organizational culture and its implications for infection prevention and control in healthcare institutions,” *Journal of Hospital Infection*, 2014, 86 (1), 1–6.
- Cavaliere, Marina, Stefania Fontana, Calogero Guccio, Domenico Lisi, Marco Ferdinando Martorana, Giacomo Pignataro, and Domenica Romeo**, “Measuring hospital care resilience: a systematic literature review,” *The European Journal of Health Economics*, 2025, pp. 1–37.
- Chandra, Amitabh, Amy Finkelstein, Adam Sacarny, and Chad Syverson**, “Health care exceptionalism? Performance and allocation in the US health care sector,” *American Economic Review*, 2016, 106 (8), 2110–2144.
- Cortes, Jorge Alberto, Martha Carolina Valderrama-Rios, Laura Cristina Nocua-Báez, Lina María Quitián, Fabio Alexander Lozada, and Giancarlo Buitrago**, “Effect of bloodstream infection on survival in COVID-19 patients admitted to an intensive care unit in Colombia: a matched cohort analysis,” *Infection Prevention in Practice*, 2023, 5 (2), 100283.
- Cronin, Christopher J and William N Evans**, “Nursing home quality, COVID-19 deaths, and excess mortality,” *Journal of Health Economics*, 2022, 82, 102592.
- Dobrzykowski, David D and Monideepa Tarafdar**, “Understanding information exchange in healthcare operations: Evidence from hospitals and patients,” *Journal of Operations Management*, 2015, 36 (1), 201–214.
- Emanuel, Natalia, Emma Harrington, and Amanda Pallais**, “The power of proximity to coworkers,” *The Quarterly Journal of Economics*, 2026, p. qjag027.
- Fakih, Mohamad G, Angelo Bufalino, Lisa Sturm, Ren-Huai Huang, Allison Ottenbacher, Karl Saake, Angela Winegar, Richard Fogel, and Joseph Cacchione**, “Coronavirus disease 2019 (COVID-19) pandemic, central-line-associated bloodstream infection (CLABSI), and catheter-associated urinary tract infection (CAUTI): the urgent need to refocus on hardwiring prevention efforts,” *Infection Control & Hospital Epidemiology*, 2022, 43 (1), 26–31.
- Fetzer, Thiemo, Christopher Rauh, and Clara Schreiner**, “The hidden toll of the pandemic: Excess mortality in non-COVID-19 hospital patients,” *Journal of Health Economics*, 2024, 95, 102882.
- Garicano, Luis**, “Hierarchies and the Organization of Knowledge in Production,” *Journal of political economy*, 2000, 108 (5), 874–904.
- Huerta, Luis E and Todd W Rice**, “Pathologic difference between sepsis and bloodstream infections,” *The journal of applied laboratory medicine*, 2019, 3 (4), 654–663.

- Khalil, Merette, Hamid Ravaghi, Dalia Samhouri, John Abo, Ahmed Ali, Hala Sakr, and Alex Camacho**, “What is “hospital resilience”? A scoping review on conceptualization, operationalization, and evaluation,” *Frontiers in public health*, 2022, *10*, 1009400.
- Lastinger, Lindsey M, Carlos R Alvarez, Aaron Kofman, Rebecca Y Konnor, David T Kuhar, Allan Nkwata, Prachi R Patel, Vaishnavi Pattabiraman, Sunny Y Xu, and Margaret A Dudeck**, “Continued increases in the incidence of healthcare-associated infection (HAI) during the second year of the coronavirus disease 2019 (COVID-19) pandemic,” *Infection Control & Hospital Epidemiology*, 2023, *44* (6), 997–1001.
- Lyons, Elizabeth**, “Team production in international labor markets: Experimental evidence from the field,” *American Economic Journal: Applied Economics*, 2017, *9* (3), 70–104.
- Miyagami, Taiju, Yuki Uehara, Taku Harada, Takashi Watari, Taro Shimizu, Ayako Nakamura, Naoya Ogura, Seiko Kushiro, Katsutoshi Masuyama, Yoshinori Kanai et al.**, “Delayed treatment of bacteremia during the COVID-19 pandemic,” *Diagnosis*, 2021, *8* (3), 327–332.
- Monegro, Alberto F, Vijayadershan Muppidi, and Hariharan Regunath**, “Hospital-acquired infections,” 2017.
- Ong, Mei-Sing, Farah Magrabi, Jeffrey Post, Sarah Morris, Johanna Westbrook, Wayne Wobcke, Ross Calcroft, and Enrico Coiera**, “Communication interventions to improve adherence to infection control precautions: a randomised crossover trial,” *BMC infectious Diseases*, 2013, *13*, 1–9.
- Ramphal, Lilly, Sumhiro Suzuki, Izah Mercy McCracken, and Amanda Addai**, “Improving hospital staff compliance with environmental cleaning behavior,” in “Baylor University Medical Center Proceedings,” Vol. 27 Taylor & Francis 2014, pp. 88–91.
- Sacchetti, Brianna, Justin Travis, Lisa L Steed, and Ginny Webb**, “Effects of COVID-19 on blood culture contamination at a tertiary care academic medical center,” *Microbiology Spectrum*, 2022, *10* (2), e00277–22.
- Sandvik, Jason J, Richard E Saouma, Nathan T Seegert, and Christopher T Stanton**, “Workplace knowledge flows,” *The Quarterly Journal of Economics*, 2020, *135* (3), 1635–1680.
- Stone, Patricia W, Cathy Mooney-Kane, Elaine L Larson, Teresa Horan, Laurent G Glance, Jack Zwanziger, and Andrew W Dick**, “Nurse working conditions and patient safety outcomes,” *Medical care*, 2007, *45* (6), 571–578.
- Upadhyay, Soumya and Dean G Smith**, “Healthcare associated infections, nurse staffing, and financial performance,” *INQUIRY: The Journal of Health Care Organization, Provision, and Financing*, 2023, *60*, 00469580231159315.

- van Buijtene, Adriana and Dona Foster**, “Does a hospital culture influence adherence to infection prevention and control and rates of healthcare associated infection? A literature review,” *Journal of infection prevention*, 2019, *20* (1), 5–17.
- von Knorring, Mia, Peter Griffiths, Jane Ball, Sara Runesdotter, and Rikard Lindqvist**, “Patient experience of communication consistency amongst staff is related to nurse–physician teamwork in hospitals,” *Nursing Open*, 2020, *7* (2), 613–617.
- Walsh, Thomas J, Rick A Bright, Aparna Ahuja, Matthew W McCarthy, Richard A Marfuggi, and Steven Q Simpson**, “Meeting the challenges of sepsis in severe coronavirus disease 2019: a call to arms,” in “Open Forum Infectious Diseases,” Vol. 10 Oxford University Press US 2023, p. ofac645.
- Warstadt, Nicholus Michael, J Reed Caldwell, Nicole Tang, Staci Mandola, Catherine Jamin, and Cassidy Dahn**, “Quality initiative to improve emergency department sepsis bundle compliance through utilisation of an electronic health record tool,” *BMJ open quality*, 2022, *11* (1).
- Weiner-Lastinger, Lindsey M, Vaishnavi Pattabiraman, Rebecca Y Konnor, Prachi R Patel, Emily Wong, Sunny Y Xu, Brittany Smith, Jonathan R Edwards, and Margaret A Dudeck**, “The impact of coronavirus disease 2019 (COVID-19) on healthcare-associated infections in 2020: A summary of data reported to the National Healthcare Safety Network,” *Infection Control & Hospital Epidemiology*, 2022, *43* (1), 12–25.
- Zingg, Walter, Alison Holmes, Markus Dettenkofer, Tim Goetting, Federica Secci, Lauren Clack, Benedetta Allegranzi, Anna-Pelagia Magiorakos, and Didier Pittet**, “Hospital organisation, management, and structure for prevention of health-care-associated infection: a systematic review and expert consensus,” *The Lancet Infectious Diseases*, 2015, *15* (2), 212–224.

7 Appendix

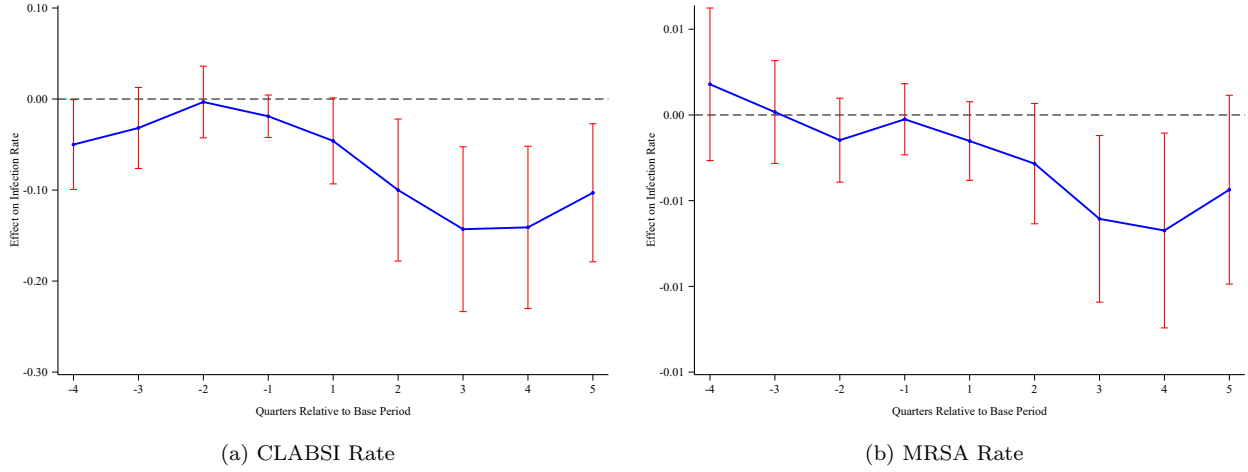


Figure 1: Pre-parallel Trend Assumption Tests for Bloodstream Infection Rates

Note: Due to inconsistent reporting during the COVID-19 period, the infection rate data for periods -4, -3, -2, and -1 correspond to the following timeframes: 2018.01–2018.12, 2018.04–2019.03, 2018.07–2019.06, and 2018.10–2019.09, respectively. The base period (0) is 2018.12–2019.12. Period 1 corresponds to 2019.04–2020.09, period 2 to 2019.07–2020.12, period 3 to 2019.10–2021.03, and period 4 to 2020.10–2021.09. Period 5 covers 2021.01–2021.12.

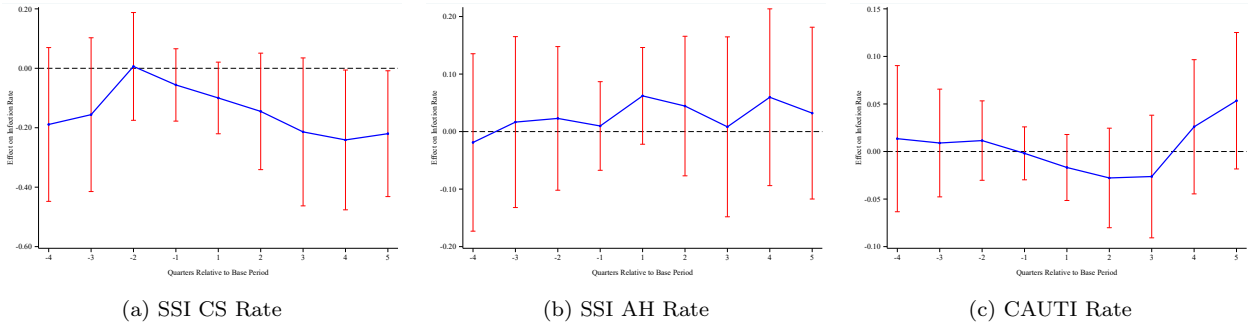


Figure 2: Pre-parallel Trend Assumption Tests for Non-Bloodstream Infection Rates

Note: Due to inconsistent reporting during the COVID-19 period, the infection rate data for periods -4, -3, -2, and -1 correspond to the following timeframes: 2018.01–2018.12, 2018.04–2019.03, 2018.07–2019.06, and 2018.10–2019.09, respectively. The base period (0) is 2018.12–2019.12. Period 1 corresponds to 2019.04–2020.09, period 2 to 2019.07–2020.12, period 3 to 2019.10–2021.03, and period 4 to 2020.10–2021.09. Period 5 covers 2021.01–2021.12.

Table 1: Variables Definitions

Name	Definition
CLABSI Rate	Central line-associated bloodstream infection (CLABSI) rate. The rate equals the number of CLABSI events divided by central-line device days, multiplied by 1,000.
CLABSI Number	Number of reported central line-associated bloodstream infection events.
CAUTI Rate	Catheter-associated urinary tract infection (CAUTI) rate. The rate equals the number of CAUTI events divided by urinary-catheter device days, multiplied by 1,000.
CAUTI Number	Number of reported catheter-associated urinary tract infection events.
SSI CS Rate	Surgical site infection rate following colon surgery. The rate equals the number of reported surgical site infections following colon surgery divided by the number of colon-surgery procedures, multiplied by 100.
SSI CS Number	Number of reported surgical site infections following colon surgery.
SSI AH Rate	Surgical site infection rate following abdominal hysterectomy. The rate equals the number of reported surgical site infections following abdominal hysterectomy divided by the number of abdominal-hysterectomy procedures, multiplied by 100.
SSI AH Number	Number of reported surgical site infections following abdominal hysterectomy.
MRSA Rate	Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) bloodstream infection rate. The rate equals the number of reported MRSA bloodstream infection events divided by patient days, multiplied by 1,000.
MRSA Number	Number of reported MRSA bloodstream infection events.
Communication.2019	Pre-pandemic hospital communication quality. It is the average of Nurse Communication, Doctor Communication, Staff Responsiveness, and Communication about Medicines, each measured on a 1-5 HCAHPS star scale. For each hospital, the measure uses the latest available pre-pandemic HCAHPS observation from 2018-2019 and is fixed throughout the analysis.
Nurse Communication	HCAHPS Nurse Communication star rating on a 1-5 scale. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Doctor Communication	HCAHPS Doctor Communication star rating on a 1-5 scale. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Staff Responsiveness	HCAHPS Staff Responsiveness star rating on a 1-5 scale. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Communication about Medicines	HCAHPS Communication about Medicines star rating on a 1-5 scale. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Clean	HCAHPS Cleanliness star rating on a 1-5 scale. It measures patients' reports of whether their room and bathroom were clean. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Recommend	HCAHPS willingness-to-recommend star rating on a 1-5 scale. It measures patients' willingness to recommend the hospital. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Quiet	HCAHPS Quietness star rating on a 1-5 scale. It measures patients' reports of whether the area around their room was quiet at night. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
High Rating	HCAHPS hospital-rating star measure on a 1-5 scale, based on patients' overall ratings of the hospital. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Hospital Summary Rating	Overall HCAHPS hospital summary star rating on a 1-5 scale. For each hospital, the measure uses the same latest available pre-pandemic HCAHPS observation from 2018-2019 used to construct Communication.2019.
Communication.2019.median	Indicator equal to one if Communication.2019 is above 3.25 and zero otherwise.
Total Cost	Hospital total cost reported in HCRIS. The baseline value is matched by hospital and year and corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Total Operating Expenses	Hospital total operating expenses reported in HCRIS. The baseline value is matched by hospital and year and corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Inpatient Hospital Revenue	Hospital inpatient revenue reported in HCRIS. The baseline value is matched by hospital and year and corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Inpatient Total Patient Revenue	Total inpatient patient revenue reported in HCRIS. The baseline value is matched by hospital and year and corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Bed Days Available	Number of available bed days reported in HCRIS for the hospital-year. The baseline value corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Total Beds	Hospital bed capacity reported in HCRIS. The baseline value corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Inpatient Bed Days Utilized	Number of inpatient bed days reported in HCRIS for the hospital-year. The baseline value corresponds to the hospital-year associated with the selected pre-pandemic baseline observation.
Staff Number	Baseline hospital staff count reported in HCRIS. The measure is matched by hospital and year and corresponds to the hospital-year associated with the selected pre-pandemic baseline observation. It is fixed throughout the baseline analysis.
FTE.Staff	Time-varying hospital staffing measure based on the number of full-time employees and contractors reported for the hospital-year. The measure varies annually and is used in the staffing alternative-explanation analysis.
Post	Indicator equal to one for infection measurement windows ending after March 2020 and zero otherwise.
Covid New Cases	County-quarter new COVID-19 cases. Quarterly new cases are constructed as changes in cumulative county cases between consecutive quarter-end observations and are matched to hospitals using county FIPS codes derived from hospital ZIP codes.
Covid Deaths	County-quarter new COVID-19 deaths. Quarterly new deaths are constructed as changes in cumulative county deaths between consecutive quarter-end observations and are matched to hospitals using county FIPS codes derived from hospital ZIP codes.
Covid Intensity	County-quarter COVID-19 intensity index constructed as the equally weighted average of the normalized ranks of quarterly new COVID-19 cases and deaths, with both components scaled to the [0, 1] interval. Higher values indicate greater local COVID-19 intensity.
SEP-1	Hospital SEP-1 performance measure for severe sepsis and septic shock. It records the proportion of eligible patients who received all required SEP-1 bundle components and ranges from 0 to 1.
Low-Volume	Indicator equal to one if a hospital has at least one missing SEP-1 report in the 2018-2019 pre-pandemic baseline sample and zero otherwise. SEP-1 nonreporting is used as an imperfect proxy for lower pre-pandemic bacterial sepsis volume and less prior BSL-related experience.
Readmission30.2019	Hospital-wide 30-day unplanned readmission rate. The measure is matched by hospital and year and is fixed using the hospital-year associated with the selected pre-pandemic baseline observation.

Table 2: Summary Statistics

Metrics	All Sample				2018		2019		2020		2021	
	Mean	Std	Max	Obs	Mean	Std	Mean	Std	Mean	Std	Mean	Std
CLABI Rate (per 1,000 device days)	0.60	0.74	3.84	30949	0.52	0.67	0.49	0.65	0.61	0.75	0.76	0.84
CAUTI Rate (per 1,000 device days)	0.69	0.74	3.65	31373	0.69	0.75	0.66	0.71	0.66	0.73	0.74	0.77
SSICS Rate (per 100 surgeries)	2.05	2.81	16.67	28976	2.12	2.83	2.08	2.81	2.06	2.86	1.99	2.77
SSIAH Rate (per 100 surgeries)	0.70	1.48	9.09	27266	0.69	1.57	0.69	1.47	0.69	1.43	0.72	1.50
MRSA Rate (per 1,000 patient days)	0.04	0.06	0.44	31710	0.04	0.05	0.04	0.05	0.04	0.05	0.05	0.06
Clean (1–5 rating)	2.89	0.97	5.00	32715	2.89	0.97	2.89	0.97	2.89	0.97	2.89	0.97
Quiet (1–5 rating)	3.20	1.19	5.00	32715	3.20	1.19	3.20	1.19	3.20	1.19	3.20	1.19
Recommend (1–5 rating)	3.18	0.99	5.00	32715	3.18	0.99	3.18	0.99	3.18	0.99	3.18	0.99
Communication (1–5 rating)	3.23	0.89	5.00	32715	3.23	0.89	3.23	0.89	3.23	0.89	3.24	0.88
High Rating (1–5 rating)	3.29	0.92	5.00	32715	3.29	0.92	3.29	0.92	3.29	0.92	3.30	0.92
Hospital Summary Rating (1–5 rating)	3.23	0.91	5.00	32715	3.23	0.91	3.23	0.91	3.23	0.91	3.23	0.91
Total Cost (billion USD)	0.29	0.45	6.16	32715	0.29	0.44	0.29	0.44	0.29	0.45	0.30	0.45
Total Operating Expenses (billion USD)	0.29	0.44	6.41	32715	0.29	0.44	0.29	0.44	0.29	0.44	0.29	0.44
Inpatient Hospital Revenue (billion USD)	0.12	0.29	7.80	32715	0.12	0.28	0.12	0.28	0.12	0.29	0.12	0.29
Staff Number (per 1,000 people)	1.28	1.83	27.42	32715	1.28	1.82	1.28	1.82	1.28	1.82	1.29	1.83
FTE Staff (per 1,000 people)	0.13	0.19	2.79	32616	0.13	0.18	0.13	0.19	0.13	0.19	0.13	0.19
Nurse Communication (1–5 rating)	3.50	0.93	5.00	32715	3.50	0.93	3.50	0.93	3.50	0.93	3.50	0.93
Doctor Communication (1–5 rating)	3.09	1.04	5.00	32715	3.09	1.04	3.09	1.04	3.09	1.04	3.09	1.04
Staff Communication (1–5 rating)	3.28	1.04	5.00	32715	3.28	1.04	3.28	1.04	3.28	1.04	3.28	1.04
Medicine Communication (1–5 rating)	3.07	1.01	5.00	32715	3.07	1.01	3.07	1.01	3.07	1.01	3.07	1.00
Covid New Cases (per 100,000 people)	0.12	0.42	5.01	32715	0.00	0.00	0.00	0.00	0.22	0.62	0.24	0.55
Covid New Deaths (per 1,000 people)	0.15	0.72	12.80	32715	0.00	0.00	0.00	0.00	0.25	0.63	0.33	1.19
Covid Intensity (0–1 rank)	0.500	0.267	0.998	32715	0.257	0.000	0.257	0.000	0.727	0.163	0.755	0.149
SEP-1 (rate 0–1)	0.17	0.38	1.00	32715	0.17	0.38	0.17	0.38	0.17	0.38	0.17	0.38
Readmission30 (rate 0–1)	0.15	0.01	0.20	32665	0.15	0.01	0.15	0.01	0.15	0.01	0.15	0.01

Note: This table reports summary statistics for key hospital-level variables from 2018 to 2021. For the full sample, the mean, standard deviation, maximum value, and number of non-missing observations are reported; for each year, the mean and standard deviation are reported. Communication, Clean, Quiet, Recommend, High Rating, Hospital Summary Rating, Nurse Communication, Doctor Communication, Staff Responsiveness, and Communication about Medicines are time-invariant baseline HCAHPS measures constructed using each hospital’s latest available pre-pandemic observation from 2018–2019. Baseline HCRIS measures, including Total Cost, Total Operating Expenses, Inpatient Hospital Revenue, and Staff Number, are matched to the hospital-year corresponding to the selected baseline observation and are fixed throughout the analysis. SEP-1 and Readmission30 are also fixed using the selected pre-pandemic baseline hospital information. In contrast, infection outcomes, FTE Staff, and COVID-19 measures vary over time and across hospitals. Infection measures follow CDC definitions. Patient-experience variables are recorded on a 1–5 scale, financial variables are expressed in billions of USD, and Staff Number and FTE Staff are expressed in thousands. Covid New Cases are reported per 100,000 people, and Covid New Deaths are reported per 1,000 people. Covid Intensity is a county-level composite index constructed as the average of the normalized ranks of Covid New Cases and Covid New Deaths. SEP-1 and Readmission30 are rates bounded between 0 and 1. CLABSI, CAUTI, SSI-CS, and SSI-AH are winsorized at the 99th percentile, while MRSA is winsorized at the 99.9th percentile.

Table 3: **Dropped Hospital Summary Statistics**

Metrics	Mean	Std
CLABSI Rate (per 1,000 device days)	0.38	2.49
CAUTI Rate (per 1,000 device days)	0.86	4.64
SSI-CS Rate (per 100 surgeries)	2.70	10.52
SSI-AH Rate (per 100 surgeries)	1.17	7.03
MRSA Rate (per 1,000 patient days)	0.01	0.08
Clean (1–5 rating)	3.59	0.97
Quiet (1–5 rating)	3.49	1.12
Recommend (1–5 rating)	3.67	0.97
Communication (1–5 rating)	3.71	0.85
High Rating (1–5 rating)	3.73	0.92
Hospital Summary Rating (1–5 rating)	3.66	0.83
Total Cost (billion USD)	0.02	0.19
Total Operating Expenses (billion USD)	0.06	0.24
Inpatient Hospital Revenue (billion USD)	0.01	0.08
Staff Number (per 1,000 people)	0.29	0.94

Note: This table reports the mean and standard deviation of selected characteristics for hospitals excluded from the main analysis sample. The reported infection outcomes include CLABSI Rate, CAUTI Rate, SSI-CS Rate, SSI-AH Rate, and MRSA Rate. Infection measures follow CDC definitions and are expressed in the units indicated in the table. Patient-experience measures are recorded on a 1–5 scale. Financial variables are expressed in billions of USD, and Staff Number is expressed in thousands. Baseline HCAHPS measures use each hospital’s latest available pre-pandemic observation from 2018–2019, while baseline HCRIS measures are matched to the corresponding hospital-year. CLABSI, CAUTI, SSI-CS, and SSI-AH are winsorized at the 99th percentile, while MRSA is winsorized at the 99.9th percentile.

Table 4: Continuous DID Results for Hospital Infection Rates

VARIABLES	BSI Rate		Non-BSI Rate		
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post $\times$ Communication_2019	-0.063*** (0.02)	-0.004*** (0.00)	-0.004 (0.02)	0.016 (0.06)	0.040 (0.04)
Post	-	-	-	-	-
Communication_2019	-0.192*** (0.03)	-0.006*** (0.00)	-0.090*** (0.03)	-0.042 (0.09)	-0.082 (0.06)
Hospital Summary Rating	0.044* (0.03)	-0.003* (0.00)	0.021 (0.02)	0.086 (0.10)	0.026 (0.06)
Clean	-0.077*** (0.01)	-0.004*** (0.00)	-0.051*** (0.01)	-0.125** (0.06)	-0.030 (0.03)
Quite	-0.005 (0.01)	-0.001 (0.00)	-0.047*** (0.01)	-0.059 (0.05)	-0.034 (0.03)
Recommend	0.020 (0.02)	-0.000 (0.00)	0.046*** (0.01)	-0.081 (0.06)	-0.024 (0.04)
High Rating	0.040* (0.02)	0.001 (0.00)	0.029 (0.02)	0.090 (0.11)	0.040 (0.05)
Total Cost	-0.059 (0.33)	0.001 (0.03)	0.564 (0.63)	-1.224 (1.10)	-0.036 (0.41)
Total Operating Expenses	0.014 (0.37)	0.004 (0.03)	-0.468 (0.64)	1.618 (1.19)	0.088 (0.43)
Inpatient Hospital Revenue	-0.070** (0.03)	-0.004 (0.00)	-0.054 (0.09)	-0.339* (0.20)	-0.051 (0.04)
Staff Number	0.059*** (0.02)	0.004*** (0.00)	0.054*** (0.02)	0.075 (0.05)	0.040* (0.02)
Covid Cases	0.012 (0.02)	-0.000 (0.00)	0.017 (0.03)	-0.162* (0.08)	0.032 (0.02)
Covid Deaths	0.018 (0.01)	0.002* (0.00)	0.017* (0.01)	0.014 (0.01)	-0.020** (0.01)
Constant	1.154*** (0.04)	0.081*** (0.00)	0.866*** (0.06)	2.213*** (0.18)	0.874*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.133	0.154	0.091	0.025	0.021
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include central line-associated bloodstream infection (CLABSI) and methicillin-resistant *Staphylococcus aureus* (MRSA) bloodstream infection. Non-BSI outcomes include catheter-associated urinary tract infection (CAUTI) and surgical site infections following colon surgery (SSI-CS) and abdominal hysterectomy (SSI-AH). CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise. Communication_2019 is the continuous measure of pre-pandemic hospital communication quality on a 1–5 scale, constructed using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Median-Split DID Results for Hospital Infection Rates

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post × Communication_2019_median	-0.086*** (0.03)	-0.004** (0.00)	-0.005 (0.02)	-0.105 (0.09)	0.035 (0.06)
Post	-	-	-	-	-
Communication_2019_median	-0.055* (0.03)	-0.003* (0.00)	-0.036 (0.03)	0.024 (0.10)	-0.054 (0.05)
Hospital Summary Rating	-0.069*** (0.02)	-0.006*** (0.00)	-0.026 (0.02)	0.074 (0.08)	-0.001 (0.05)
Clean	-0.090*** (0.01)	-0.004*** (0.00)	-0.056*** (0.01)	-0.127** (0.06)	-0.034 (0.03)
Quite	-0.007 (0.01)	-0.001 (0.00)	-0.048*** (0.01)	-0.059 (0.05)	-0.034 (0.03)
Recommend	0.017 (0.02)	-0.000 (0.00)	0.044*** (0.01)	-0.081 (0.06)	-0.025 (0.03)
High Rating	0.008 (0.02)	-0.000 (0.00)	0.017 (0.02)	0.085 (0.10)	0.032 (0.05)
Total Cost	0.034 (0.34)	0.004 (0.03)	0.604 (0.63)	-1.198 (1.10)	-0.006 (0.41)
Total Operating Expenses	-0.093 (0.38)	0.001 (0.03)	-0.512 (0.64)	1.589 (1.19)	0.054 (0.43)
Inpatient Hospital Revenue	-0.059 (0.04)	-0.003 (0.00)	-0.050 (0.09)	-0.337* (0.20)	-0.047 (0.04)
Staff Number	0.064*** (0.02)	0.004*** (0.00)	0.057*** (0.02)	0.075 (0.05)	0.041* (0.02)
Covid Cases	0.023 (0.02)	0.000 (0.00)	0.020 (0.03)	-0.174** (0.08)	0.029 (0.02)
Covid Deaths	0.016 (0.01)	0.002* (0.00)	0.016* (0.01)	0.014 (0.01)	-0.020** (0.01)
Constant	0.987*** (0.04)	0.073*** (0.00)	0.800*** (0.06)	2.179*** (0.19)	0.819*** (0.09)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.125	0.153	0.090	0.025	0.020
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise.

Communication_2019_median equals one for hospitals with above-median pre-pandemic hospital communication quality and zero otherwise. The underlying communication measure uses each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: DID Results with Hospital Fixed Effects

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post $\times$ Communication_2019	-0.067*** (0.02)	-0.004*** (0.00)	-0.011 (0.02)	0.029 (0.06)	0.044 (0.03)
Post	-	-	-	-	-
Communication_2019	-	-	-	-	-
Hospital Summary Rating	-	-	-	-	-
Clean	-	-	-	-	-
Quite	-	-	-	-	-
Recommend	-	-	-	-	-
High Rating	-	-	-	-	-
Total Cost	-	-	-	-	-
Total Operating Expenses	-	-	-	-	-
Inpatient Hospital Revenue	-	-	-	-	-
Staff Number	-	-	-	-	-
Covid Cases	-0.035 (0.02)	-0.001 (0.00)	-0.003 (0.02)	-0.068* (0.03)	0.030 (0.03)
Covid Deaths	0.023** (0.01)	0.002*** (0.00)	0.017*** (0.01)	0.013 (0.01)	-0.021** (0.01)
Constant	0.705*** (0.03)	0.047*** (0.00)	0.701*** (0.03)	2.016*** (0.09)	0.630*** (0.05)
Hospital Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,943	31,705	31,368	28,971	27,252
R-squared	0.525	0.520	0.515	0.446	0.411
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise. Communication_2019 is constructed using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019 and is fixed throughout the analysis. Post equals one for measurement windows ending after March 2020 and zero otherwise. All specifications include hospital and year-quarter fixed effects. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: DID Results Controlling for Pre-Pandemic Readmission Rates

VARIABLES	BSI Rate		CAUTI Rate	Non-BSI Rate	
	CLABI Rate	MRSA Rate		SSI CS Rate	SSI AH Rate
Post × Communication_2019	-0.070*** (0.02)	-0.004*** (0.00)	-0.015 (0.02)	0.029 (0.06)	0.032 (0.04)
Post × Readmission30_2019	-1.439 (1.73)	-0.040 (0.13)	-1.847 (1.19)	-0.049 (6.24)	-5.484* (2.81)
Post	-	-	-	-	-
Communication_2019	-	-	-	-	-
Readmission30_2019	-	-	-	-	-
Hospital Summary Rating	-	-	-	-	-
Clean	-	-	-	-	-
Quite	-	-	-	-	-
Recommend	-	-	-	-	-
High Rating	-	-	-	-	-
Total Cost	-	-	-	-	-
Total Operating Expenses	-	-	-	-	-
Inpatient Hospital Revenue	-	-	-	-	-
Staff Number	-	-	-	-	-
Covid Cases	-0.034 (0.02)	-0.001 (0.00)	-0.002 (0.02)	-0.068* (0.04)	0.033 (0.03)
Covid Deaths	0.023** (0.01)	0.002*** (0.00)	0.017*** (0.01)	0.013 (0.01)	-0.021** (0.01)
Constant	0.821*** (0.13)	0.050*** (0.01)	0.850*** (0.10)	2.021*** (0.52)	1.068*** (0.25)
Hospital Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,915	31,656	31,345	28,963	27,208
R-squared	0.525	0.520	0.515	0.446	0.411
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise. Communication_2019 is constructed using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Readmission30_2019 is the hospital-wide 30-day readmission rate associated with the selected pre-pandemic baseline hospital-year. Both measures are fixed before COVID-19 and are interacted with Post. Post equals one for measurement windows ending after March 2020 and zero otherwise. All specifications include hospital and year-quarter fixed effects. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table 8: DID Results Controlling for Hospital Staffing

VARIABLES	BSI Rate		CAUTI Rate	Non-BSI Rate	
	CLABI Rate	MRSA Rate		SSI CS Rate	SSI AH Rate
Post × Communication_2019	-0.067*** (0.02)	-0.004*** (0.00)	-0.011 (0.02)	0.020 (0.06)	0.042 (0.03)
Post × FTE_Staff	-0.023 (0.04)	0.000 (0.00)	0.013 (0.04)	-0.093 (0.17)	0.027 (0.08)
FTE_Staff	-0.147 (0.27)	-0.035 (0.03)	0.098 (0.31)	2.400 (2.57)	0.481 (0.43)
Post	-	-	-	-	-
Communication_2019	-	-	-	-	-
Hospital Summary Rating	-	-	-	-	-
Clean	-	-	-	-	-
Quite	-	-	-	-	-
Recommend	-	-	-	-	-
High Rating	-	-	-	-	-
Total Cost	-	-	-	-	-
Total Operating Expenses	-	-	-	-	-
Inpatient Hospital Revenue	-	-	-	-	-
Staff Number	-	-	-	-	-
Covid Cases	-0.035 (0.02)	-0.001 (0.00)	-0.003 (0.02)	-0.070* (0.04)	0.028 (0.03)
Covid Deaths	0.023** (0.01)	0.002*** (0.00)	0.017*** (0.01)	0.014 (0.01)	-0.021** (0.01)
Constant	0.726*** (0.05)	0.051*** (0.00)	0.685*** (0.05)	1.690*** (0.36)	0.558*** (0.06)
Hospital Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,853	31,612	31,277	28,879	27,167
R-squared	0.525	0.521	0.515	0.447	0.412
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise. Communication_2019 is constructed using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019 and is fixed throughout the analysis. FTE_Staff is the time-varying hospital staffing measure and is observed annually. The specification includes FTE_Staff and its interaction with Post. Post equals one for measurement windows ending after March 2020 and zero otherwise. All specifications include hospital and year-quarter fixed effects. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table 9: Triple-Difference Results by Pre-Pandemic Sepsis Volume

VARIABLES	BSI Rate		CAUTI Rate	Non-BSI Rate	
	CLABI Rate	MRSA Rate		SSI CS Rate	SSI AH Rate
Post $\times$ Communication_2019 $\times$ Low_Volume	-0.005 (0.05)	-0.008** (0.00)	-0.014 (0.05)	-0.037 (0.23)	0.276* (0.15)
Post $\times$ Communication_2019	-0.045* (0.02)	-0.001 (0.00)	0.015 (0.02)	0.095 (0.06)	0.048 (0.04)
Low_Volume $\times$ Post	-0.106 (0.18)	0.025* (0.01)	-0.079 (0.20)	-0.357 (0.88)	-1.333** (0.64)
Communication_2019 $\times$ Low_Volume	-	-	-	-	-
Post	-	-	-	-	-
Communication_2019	-	-	-	-	-
Low_Volume	-	-	-	-	-
Hospital Summary Rating	-	-	-	-	-
Clean	-	-	-	-	-
Quite	-	-	-	-	-
Recommend	-	-	-	-	-
High Rating	-	-	-	-	-
Total Cost	-	-	-	-	-
Total Operating Expenses	-	-	-	-	-
Inpatient Hospital Revenue	-	-	-	-	-
Staff Number	-	-	-	-	-
Covid Cases	-0.034 (0.02)	-0.001 (0.00)	-0.002 (0.02)	-0.067* (0.03)	0.030 (0.03)
Covid Deaths	0.023** (0.01)	0.002*** (0.00)	0.017*** (0.01)	0.013 (0.01)	-0.021** (0.01)
Constant	0.678*** (0.04)	0.044*** (0.00)	0.669*** (0.03)	1.938*** (0.09)	0.632*** (0.06)
Hospital Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,943	31,705	31,368	28,971	27,252
R-squared	0.526	0.520	0.515	0.447	0.412
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: This table reports triple-difference estimates for hospital infection rates. BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise. Low_Volume equals one if a hospital has at least one missing SEP-1 report in the 2018–2019 pre-pandemic baseline sample and zero otherwise. SEP-1 nonreporting is used as an imperfect proxy for lower pre-pandemic bacterial sepsis volume and less prior BSI-related experience. Communication_2019 is constructed using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. All specifications include hospital and year-quarter fixed effects. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: **Hospital Infection Rates and COVID-19 Intensity**

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Covid Intensity $\times$ Communication_2019_median	-0.120* (0.07)	-0.009** (0.00)	0.007 (0.04)	-0.116 (0.15)	0.051 (0.10)
Covid Intensity	0.275*** (0.06)	0.007 (0.01)	0.119 (0.07)	-0.134 (0.26)	-0.028 (0.15)
Communication_2019_median	-	-	-	-	-
Hospital Summary Rating	-	-	-	-	-
Clean	-	-	-	-	-
Quite	-	-	-	-	-
Recommend	-	-	-	-	-
High Rating	-	-	-	-	-
Total Cost	-	-	-	-	-
Total Operating Expenses	-	-	-	-	-
Inpatient Hospital Revenue	-	-	-	-	-
Staff Number	-	-	-	-	-
Constant	0.483*** (0.04)	0.039*** (0.00)	0.625*** (0.04)	2.143*** (0.14)	0.703*** (0.08)
Hospital Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,943	31,705	31,368	28,971	27,252
R-squared	0.525	0.519	0.514	0.446	0.411
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise.

Communication_2019_median equals 1 for hospitals with above-median pre-pandemic hospital communication quality and 0 otherwise. The underlying communication measure uses each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Covid Intensity is the equally weighted average of the normalized ranks of county-level quarterly new COVID-19 cases and deaths, with each component scaled to the $[0, 1]$ interval. The interaction allows the relationship between local COVID-19 intensity and infection rates to differ across the two communication groups. All specifications include hospital and year-quarter fixed effects. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: DID Results Controlling for Pre-Pandemic Hospital Quietness

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post × Communication_2019	-0.084*** (0.03)	-0.006*** (0.00)	-0.003 (0.02)	-0.021 (0.07)	0.061 (0.04)
Post × Quite	0.025 (0.02)	0.003** (0.00)	-0.002 (0.02)	0.045 (0.04)	-0.025 (0.03)
Post	-	-	-	-	-
Communication_2019	-0.182*** (0.03)	-0.004** (0.00)	-0.091*** (0.03)	-0.023 (0.09)	-0.093 (0.06)
Quite	-0.018 (0.01)	-0.003** (0.00)	-0.046** (0.02)	-0.081 (0.05)	-0.022 (0.03)
Clean	-0.077*** (0.01)	-0.004*** (0.00)	-0.051*** (0.01)	-0.125** (0.06)	-0.030 (0.03)
Hospital Summary Rating	0.044* (0.03)	-0.003* (0.00)	0.021 (0.02)	0.087 (0.10)	0.026 (0.06)
Recommend	0.020 (0.02)	-0.000 (0.00)	0.046*** (0.01)	-0.081 (0.06)	-0.024 (0.04)
High Rating	0.040* (0.02)	0.001 (0.00)	0.029 (0.02)	0.089 (0.11)	0.040 (0.05)
Total Cost	-0.060 (0.33)	0.001 (0.03)	0.564 (0.63)	-1.227 (1.10)	-0.034 (0.41)
Total Operating Expenses	0.015 (0.37)	0.004 (0.03)	-0.468 (0.64)	1.621 (1.19)	0.086 (0.43)
Inpatient Hospital Revenue	-0.070** (0.03)	-0.004 (0.00)	-0.054 (0.09)	-0.340* (0.20)	-0.051 (0.04)
Staff Number	0.059*** (0.02)	0.004*** (0.00)	0.054*** (0.02)	0.075 (0.05)	0.040* (0.02)
Covid Cases	0.014 (0.02)	0.000 (0.00)	0.016 (0.03)	-0.158* (0.08)	0.030 (0.02)
Covid Deaths	0.018 (0.01)	0.002* (0.00)	0.017* (0.01)	0.014 (0.01)	-0.020** (0.01)
Constant	1.153*** (0.04)	0.081*** (0.00)	0.866*** (0.06)	2.213*** (0.18)	0.874*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.133	0.155	0.091	0.025	0.021
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. CLABSI and MRSA are classified as BSI outcomes with higher COVID-induced diagnostic noise, while CAUTI, SSI-CS, and SSI-AH are classified as non-BSI outcomes with lower COVID-induced diagnostic noise. Communication_2019 and Quiet are both HCAHPS measures constructed using the same latest available pre-pandemic observation from 2018–2019 for each hospital. Both are measured on a 1–5 scale and interacted with Post. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table 12: DID Results Using Nurse Communication

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post × Nurse Communication	-0.052*** (0.02)	-0.004*** (0.00)	0.013 (0.01)	0.030 (0.05)	0.020 (0.03)
Post	-	-	-	-	-
Nurse Communication	-0.034 (0.02)	0.002* (0.00)	-0.037* (0.02)	0.099 (0.08)	-0.002 (0.05)
Hospital Summary Rating	-0.069*** (0.02)	-0.009*** (0.00)	-0.022 (0.02)	0.001 (0.09)	-0.019 (0.05)
Clean	-0.090*** (0.01)	-0.004*** (0.00)	-0.056*** (0.01)	-0.134** (0.05)	-0.035 (0.03)
Quite	-0.007 (0.01)	-0.001 (0.00)	-0.048*** (0.01)	-0.061 (0.05)	-0.035 (0.03)
Recommend	0.020 (0.02)	-0.000 (0.00)	0.046*** (0.01)	-0.087 (0.06)	-0.025 (0.03)
High Rating	0.019 (0.02)	-0.000 (0.00)	0.022 (0.02)	0.060 (0.11)	0.029 (0.05)
Total Cost	0.011 (0.32)	0.006 (0.03)	0.585 (0.62)	-1.122 (1.09)	0.004 (0.41)
Total Operating Expenses	-0.065 (0.36)	-0.001 (0.03)	-0.492 (0.63)	1.506 (1.18)	0.044 (0.43)
Inpatient Hospital Revenue	-0.068* (0.04)	-0.004 (0.00)	-0.053 (0.09)	-0.330 (0.20)	-0.048 (0.04)
Staff Number	0.066*** (0.02)	0.004*** (0.00)	0.057*** (0.02)	0.077 (0.05)	0.042** (0.02)
Covid Cases	0.019 (0.02)	0.000 (0.00)	0.022 (0.03)	-0.154* (0.08)	0.031 (0.02)
Covid Deaths	0.016 (0.01)	0.002* (0.00)	0.016* (0.01)	0.013 (0.01)	-0.020** (0.01)
Constant	1.107*** (0.05)	0.078*** (0.00)	0.850*** (0.06)	2.131*** (0.18)	0.845*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.125	0.153	0.090	0.025	0.020
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. Nurse Communication is measured on a 1–5 scale using each hospital’s latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table 13: DID Results Using Doctor Communication

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post $\times$ Doctor Communication	-0.047*** (0.02)	-0.002 (0.00)	-0.006 (0.01)	0.071 (0.05)	0.033 (0.03)
Post	-	-	-	-	-
Doctor Communication	-0.054*** (0.02)	-0.003*** (0.00)	-0.001 (0.02)	-0.095 (0.07)	-0.035 (0.03)
Hospital Summary Rating	-0.052** (0.02)	-0.005*** (0.00)	-0.037 (0.02)	0.104 (0.08)	-0.003 (0.05)
Clean	-0.094*** (0.01)	-0.004*** (0.00)	-0.057*** (0.01)	-0.128** (0.06)	-0.035 (0.03)
Quite	-0.003 (0.01)	-0.001 (0.00)	-0.048*** (0.01)	-0.055 (0.05)	-0.034 (0.03)
Recommend	0.020 (0.02)	-0.000 (0.00)	0.045*** (0.01)	-0.079 (0.06)	-0.024 (0.04)
High Rating	0.018 (0.02)	0.000 (0.00)	0.016 (0.02)	0.094 (0.10)	0.034 (0.05)
Total Cost	-0.006 (0.35)	0.002 (0.03)	0.607 (0.64)	-1.263 (1.08)	-0.018 (0.41)
Total Operating Expenses	-0.031 (0.39)	0.004 (0.03)	-0.515 (0.66)	1.671 (1.16)	0.071 (0.43)
Inpatient Hospital Revenue	-0.065* (0.03)	-0.004 (0.00)	-0.051 (0.09)	-0.340* (0.20)	-0.049 (0.04)
Staff Number	0.061*** (0.02)	0.004*** (0.00)	0.058*** (0.02)	0.072 (0.05)	0.041** (0.02)
Covid Cases	0.021 (0.02)	0.001 (0.00)	0.019 (0.03)	-0.149* (0.09)	0.033 (0.02)
Covid Deaths	0.016 (0.01)	0.002* (0.00)	0.016* (0.01)	0.013 (0.01)	-0.020** (0.01)
Constant	1.088*** (0.04)	0.079*** (0.00)	0.833*** (0.05)	2.211*** (0.18)	0.854*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.127	0.154	0.090	0.025	0.021
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. Doctor Communication is measured on a 1–5 scale using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: DID Results Using Staff Communication

VARIABLES	BSI Rate			Non-BSI Rate	
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post × Staff Communication	-0.047** (0.02)	-0.003*** (0.00)	-0.009 (0.01)	-0.013 (0.04)	0.032 (0.03)
Post	-	-	-	-	-
Staff Communication	-0.083*** (0.02)	-0.003** (0.00)	-0.067*** (0.02)	-0.074 (0.05)	-0.082** (0.03)
Hospital Summary Rating	-0.030 (0.02)	-0.005*** (0.00)	0.009 (0.02)	0.119 (0.09)	0.031 (0.05)
Clean	-0.077*** (0.01)	-0.003*** (0.00)	-0.047*** (0.01)	-0.115** (0.05)	-0.025 (0.03)
Quite	-0.010 (0.01)	-0.001 (0.00)	-0.049*** (0.01)	-0.061 (0.05)	-0.036 (0.03)
Recommend	0.014 (0.02)	-0.000 (0.00)	0.042*** (0.01)	-0.084 (0.06)	-0.027 (0.04)
High Rating	0.020 (0.02)	0.000 (0.00)	0.025 (0.02)	0.095 (0.11)	0.039 (0.05)
Total Cost	-0.006 (0.33)	0.003 (0.03)	0.571 (0.62)	-1.243 (1.10)	-0.040 (0.41)
Total Operating Expenses	-0.062 (0.37)	0.002 (0.03)	-0.487 (0.62)	1.627 (1.19)	0.084 (0.43)
Inpatient Hospital Revenue	-0.066* (0.04)	-0.004 (0.00)	-0.053 (0.09)	-0.340* (0.20)	-0.051 (0.04)
Staff Number	0.063*** (0.02)	0.004*** (0.00)	0.055*** (0.02)	0.073 (0.05)	0.040* (0.02)
Covid Cases	0.014 (0.02)	-0.000 (0.00)	0.014 (0.03)	-0.171** (0.08)	0.030 (0.02)
Covid Deaths	0.018 (0.01)	0.002* (0.00)	0.018* (0.01)	0.015 (0.01)	-0.020* (0.01)
Constant	1.117*** (0.04)	0.080*** (0.00)	0.863*** (0.05)	2.237*** (0.17)	0.879*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.130	0.155	0.092	0.025	0.021
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. Staff Responsiveness is measured on a 1–5 scale using each hospital’s latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table 15: DID Results Using Medicine Communication

VARIABLES	BSI Rate		CAUTI Rate	Non-BSI Rate	
	CLABI Rate	MRSA Rate		SSI CS Rate	SSI AH Rate
Post × Medicine Communication	-0.047*** (0.02)	-0.003** (0.00)	-0.008 (0.01)	-0.038 (0.05)	0.035 (0.03)
Post	-	-	-	-	-
Medicine Communication	-0.031* (0.02)	0.000 (0.00)	-0.017 (0.02)	0.039 (0.05)	-0.019 (0.03)
Hospital Summary Rating	-0.065*** (0.02)	-0.007*** (0.00)	-0.025 (0.02)	0.050 (0.09)	-0.013 (0.05)
Clean	-0.087*** (0.01)	-0.004*** (0.00)	-0.055*** (0.01)	-0.130** (0.06)	-0.035 (0.03)
Quite	-0.009 (0.01)	-0.001 (0.00)	-0.048*** (0.01)	-0.059 (0.05)	-0.035 (0.03)
Recommend	0.017 (0.02)	-0.000 (0.00)	0.045*** (0.01)	-0.081 (0.06)	-0.025 (0.04)
High Rating	0.012 (0.02)	-0.000 (0.00)	0.018 (0.02)	0.083 (0.11)	0.031 (0.05)
Total Cost	0.047 (0.34)	0.005 (0.03)	0.610 (0.63)	-1.198 (1.11)	-0.003 (0.41)
Total Operating Expenses	-0.107 (0.38)	0.000 (0.03)	-0.519 (0.64)	1.589 (1.20)	0.052 (0.43)
Inpatient Hospital Revenue	-0.063* (0.04)	-0.004 (0.00)	-0.051 (0.09)	-0.338* (0.20)	-0.049 (0.04)
Staff Number	0.066*** (0.02)	0.004*** (0.00)	0.058*** (0.02)	0.076 (0.05)	0.042** (0.02)
Covid Cases	0.023 (0.02)	0.000 (0.00)	0.018 (0.03)	-0.169** (0.08)	0.031 (0.02)
Covid Deaths	0.017 (0.01)	0.002* (0.00)	0.017* (0.01)	0.014 (0.02)	-0.020** (0.01)
Constant	1.082*** (0.04)	0.078*** (0.00)	0.837*** (0.05)	2.197*** (0.17)	0.850*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	30,949	31,710	31,373	28,976	27,266
R-squared	0.125	0.153	0.090	0.025	0.020
Sample Mean	0.62	0.04	0.69	2.07	0.71

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. Communication about Medicines is measured on a 1–5 scale using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * p<0.10, ** p<0.05, *** p<0.01.

Table 16: Subsample Check for DID Results for Hospital Infection Rates

VARIABLES	BSI Rate		Non-BSI Rate		
	CLABI Rate	MRSA Rate	CAUTI Rate	SSI CS Rate	SSI AH Rate
Post $\times$ Communication_2019	-0.063*** (0.02)	-0.004** (0.00)	-0.006 (0.02)	0.025 (0.06)	0.040 (0.03)
Post	-	-	-	-	-
Communication_2019	-0.189*** (0.03)	-0.006*** (0.00)	-0.090*** (0.03)	-0.031 (0.09)	-0.095* (0.05)
Hospital Summary Rating	0.045* (0.03)	-0.003 (0.00)	0.026 (0.02)	0.094 (0.10)	0.031 (0.06)
Clean	-0.080*** (0.01)	-0.004*** (0.00)	-0.052*** (0.01)	-0.175*** (0.06)	-0.029 (0.03)
Quite	-0.006 (0.01)	-0.001 (0.00)	-0.045*** (0.02)	-0.072 (0.05)	-0.032 (0.03)
Recommend	0.023 (0.02)	-0.000 (0.00)	0.044** (0.02)	-0.088 (0.07)	-0.024 (0.04)
High Rating	0.037 (0.02)	0.001 (0.00)	0.027 (0.02)	0.122 (0.11)	0.048 (0.05)
Total Cost	0.105 (0.39)	0.029 (0.03)	0.789 (0.68)	-0.597 (1.15)	0.316 (0.34)
Total Operating Expenses	-0.170 (0.42)	-0.027 (0.03)	-0.736 (0.70)	0.884 (1.25)	-0.362 (0.36)
Inpatient Hospital Revenue	-0.067* (0.04)	-0.003 (0.00)	-0.047 (0.09)	-0.336* (0.19)	-0.043 (0.04)
Staff Number	0.064*** (0.02)	0.004*** (0.00)	0.063*** (0.02)	0.099* (0.05)	0.063*** (0.02)
Covid Cases	0.010 (0.02)	-0.000 (0.00)	0.011 (0.03)	-0.128 (0.09)	0.040 (0.03)
Covid Deaths	0.016 (0.01)	0.002* (0.00)	0.016* (0.01)	0.012 (0.01)	-0.023* (0.01)
Constant	1.146*** (0.05)	0.080*** (0.00)	0.860*** (0.06)	2.220*** (0.18)	0.849*** (0.08)
State Fixed Effect	✓	✓	✓	✓	✓
Year Quarter Fixed Effect	✓	✓	✓	✓	✓
Cluster: State	✓	✓	✓	✓	✓
Observations	27,854	28,583	28,264	25,939	24,405
R-squared	0.140	0.159	0.089	0.029	0.021

Note: BSI outcomes include CLABSI and MRSA bloodstream infection. Non-BSI outcomes include CAUTI, SSI-CS, and SSI-AH. Communication_2019 is constructed using each hospital's latest available pre-pandemic HCAHPS observation from 2018–2019. The restricted sample retains hospitals with Communication_2019 greater than 3.2 or below 2.8 and excludes hospitals with intermediate communication scores. Post equals one for measurement windows ending after March 2020 and zero otherwise. Standard errors are clustered at the state level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.